Lecture Notes

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Contents

Part I
MATH 61CM: Course Notes

Week 1

The Real Numbers

1.1 Analysis: Mon/

1.1.1 Introduction to $\mathbb{R}$

Real analysis is built on the structure of fields — specifically the real numbers $\mathbb{R}$.

Definition 1.1 · The Three Axioms for $\mathbb{R}$

The real numbers are the **unique** field that satisfy all three of the following axioms:

1. **Arithmetic:** Field axioms.
2. **Order:** Order axioms.
3. **No Holes:** Completeness axiom.

1.1.2 Field Axioms

Definition 1.2 · Field

A set combined with two binary operations $(\mathbb{F}, +, \cdot)$ is a **field** if it satisfies the following properties:

1. F1 (Commutativity): For all $a, b \in \mathbb{F}$,

$$a + b = b + a \text{ and } ab = ba$$

2. F2 (Associativity): For all $a, b, c \in \mathbb{F}$,

$$(a + b) + c = a + (b + c) \text{ and } (ab)c = a(bc)$$

3. F3 (Distributivity): For all $a, b, c \in \mathbb{F}$,

$$a(b + c) = ab + ac$$

4. F4 (Identities Exist): There are special elements, $0, 1 \in \mathbb{F}$ s.t.

$$a + 0 = 0 + a = a \text{ and } a \cdot 1 = a \text{ for all } a \in \mathbb{F}$$

5. F5 (Additive Inverse): For every $a \in \mathbb{F}$ there exists $-a$ s.t. $a + (-a) = 0$
6. F6 (Multiplicative Inverse): For every $a \neq 0$, there exists a^{-1} s.t. $a \cdot a^{-1} = 1$

Remark. In F4, it is assumed that $1 \neq 0$.

Remark. A set combined with one binary operation that fulfills F1-F5 is called a commutative group, $(G, \circ)$.

Lemma 1.1 · Inverses are Unique

For all $a, b, c \in \mathbb{F}$, if $a + b = 0$ and $a + c = 0$, then $b = c$.

Proof. We start with $b = 0 + b$ (F4). This is equal to $b + 0$ (F2) which is equal to $b + a + c$. Now we can use Associativity which states that $b + a + c = (b + a) + c\,(= a + b + c)$ (F1). $a + b = 0$ so this is simply $0 + c = c$ (F4). We have now proved our claim that $b = c$. ∎

Lemma 1.2 · "Multiplying" by the zero element

For all $a \in \mathbb{F}, 0 \cdot a = 0$

Proof. The idea behind this proof is quite simple. Note that $0 \cdot a = (0 + 0) \cdot a = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0 = 0 \cdot a + 0 \cdot a$. If we add $-(0 \cdot a)$ to both sides, we get that $0 \cdot a - 0 \cdot a = 0 \cdot a + 0 \cdot a - 0 \cdot a$. This implies that $0 = 0 \cdot a$ ∎

Lemma 1.3 · The relationship between -1 and the additive inverse.

For all $a \in \mathbb{F}, (-1) \cdot a = -a$

Proof. Note that $a + (-1) \cdot a = 1 \cdot a + (-1) \cdot a = (1 + (-1)) \cdot a = 0 \cdot a$. Using lemma 1.2, we know that $0 \cdot a = 0$ which means that $a + (-1) \cdot a = 0$. Since inverses are unique (lemma 1.1), we know that $(-1) \cdot a = -a$. ∎

1.1.3 Order Axioms

Definition 1.3 · Order Axioms

An ordered field has a positive set $\mathbb{P}$ satisfying:

1. **Trichotomy:** For $a \in \mathbb{F}$, exactly one is true: $a = 0$, $a \in \mathbb{P}$, $-a \in \mathbb{P}$.
2. **Closure:** $\mathbb{P}$ is closed under addition and multiplication.

Remark. Here is some more common notation:

1. Write $a > 0$ for $a \in \mathbb{P}$
2. Write $a > b$ or $b < a$ if $a - b = a + (-b) > 0$
3. Write $a \geq b$ or $b \leq a$ if $(a > b$ or $a = b)$

Lemma 1.4 · 1>0

1 *is greater than* 0

Proof. Assume for the sake of contradiction that $1 < 0$ (We know that $1 \neq 0$ by definition). This implies that $(-1) > 0$ by F4 and Trichotomy. By O2, $(-1)(-1) > 0$. By Lemma 1.3, $(-1)(-1) = 1$ and thus $1 > 0$ which contradicts our initial claim. Therefore $1 > 0$. ∎

1.1.4 Completeness

Definition 1.4 · Completeness

An ordered field $(F, +, \cdot)$is **complete** if any non-empty subset $S \subseteq \mathbb{F}$ bounded above has a **least upper bound** (supremum).

Remark. For now, take $\mathbb{R}$ to be the ONLY complete ordered field.

Definition 1.5 · Bounds

$S \subseteq \mathbb{F}$ is $\left\{ \begin{array}{c} \text{bdd above} \\ \text{bdd below} \end{array} \right\}$ if $\exists \alpha \in \mathbb{F}$ s.t. $\left\{ \begin{array}{c} x \leq \alpha \\ x \geq \alpha \end{array} \right\}, \forall x \in \mathbb{F}$.

Such α is called a $\left\{ \begin{array}{c} \text{an upper bound} \\ \text{a lower bound} \end{array} \right\}$

Remark. $S \subseteq \mathbb{F}$ is bounded if it is bounded above and below.

Definition 1.6 · Supremum

Let $S \subseteq \mathbb{F}$. $\beta \in \mathbb{F}$ is a least upper bound if the following both hold:

1. β is an upper bound of S

2. If α is an upper bound of S then $\beta \leq \alpha$. Similarly for the greatest lower bound.

We denote the least upper bound as $\sup S$ and the greatest lower bound as $\inf S$.

Remark. $\sup S$ need not be in S.

Definition 1.7 · $\mathbb{N}, \mathbb{Q}, \mathbb{Z}$

1. The **natural numbers** are defined as $\mathbb{N} = \{1, 1+1, 1+1+1, \ldots\}$
2. The **integers** are defined as $\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-n : n \in \mathbb{N}\} \subseteq \mathbb{R}$
3. The **rational numbers** are defined as $\mathbb{Q} = \left\{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\right\}$

Theorem 1.5 · **Archimedean Principle**

$\mathbb{N}$ is not bounded above. Equivalently, for any two positive real numbers $a, b \in \mathbb{R}$, there exists a natural number n s.t. $a < n \cdot b$.

Proof. Assume for the sake of contradiction that $\mathbb{N}$ was bounded above. If it were, since $\mathbb{N}$ is a subset of $\mathbb{R}$, we know that there exists $\alpha = \sup \mathbb{N}$ because of completeness. $\alpha = \sup \mathbb{N} \implies n \leq \alpha \, \forall n \in \mathbb{N} \implies n + 1 \leq \alpha \implies n \leq \alpha - 1$ which contradicts our assumption that α was the sup. ■

1.1.5 Sequences

Definition 1.8 · **Sequence**

A **sequence** of real numbers is a map $\mathbb{N} \to \mathbb{R}$. We denote this by $\{a_n\}_{n=1}^{\infty}$, where $a_n = a(n)$

Definition 1.9 · **Limit of a Sequence**

Let $\{a_n\}_{n=1}^{\infty} \subseteq \mathbb{R}$. We say $\{a_n\}_{n=1}^{\infty}$ has a limit $L \in \mathbb{R}$, written $\lim_{n \to \infty} a_n = L$ if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|a_n - L| < \varepsilon$ for all $n \geq N$.

Remark. We say $\{a_n\}_{n=1}^{\infty} \subseteq \mathbb{R}$ is $\left\{ \begin{array}{c} \text{convergent if it has a limit} \\ \text{divergent otherwise} \end{array} \right\}$

Theorem 1.6 · **Monotone Convergence**

If $\{a_n\}$ is bounded above and increasing ($a_n \leq a_{n+1}$), it converges to $\sup\{a_n\}$.

Proof. Let $\alpha = \sup a_n$, which we know exists because of the completeness property of the reals. Now we will prove there exists a_j s.t. $\alpha > a_j > \alpha - \frac{1}{j}$. If such a_j didn't exist, then $\alpha - \frac{1}{j}$ would be the supremum, which is a contradiction. Therefore such a_j exists. Since $\alpha - \frac{1}{j} < a_j < \alpha$, we know that $|\alpha - a_j| < |\alpha - \frac{1}{j}|$. For all $\varepsilon > 0$, we know there exists $\frac{1}{J}$ s.t. $\frac{1}{J} < \varepsilon$ due to the Archimedean property of the real numbers. This means that for all $j \geq J$ (or $\frac{1}{j} < \frac{1}{J}$), $|a_j - \alpha| \leq \frac{1}{j} \leq \frac{1}{J} < \varepsilon$ ■

1.2 Linear Algebra: Thu/Fri

1.2.1 Vector Spaces

Definition 1.10 · Vector Space

A **vector space** V over a field $\mathbb{F}$ is a non-empty set of vectors with two operations:

$$+ : V \times V \to V$$
$$\cdot : \mathbb{F} \times V \to V$$

such that:

1. $(V, +)$ forms a commutative group:
   (A1) $\forall v_1, v_2, v_3$: $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$
   (A2) $\exists$ zero vector $\vec{0}$: $v + \vec{0} = v$
   (A3) $\forall v$, $\exists(-v)$ s.t. $v + (-v) = \vec{0}$
   (A4) $v + w = w + v$
2. $\forall v \in V$: $1 \cdot v = v$
3. $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$
4. Distributive laws: $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$ and $\alpha \cdot (v + w) = \alpha \cdot v + \alpha \cdot w$

Lemma 1.7 · Zero Times Anything

$0 \cdot v = \vec{0}$ *for all* $v \in V$. *(Same proof as in* $\mathbb{F}$*!)*

Example:

1. $\mathbb{F}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{F}\}$ with componentwise operations.
2. $\mathbb{F}^S$ = space of functions from $S \to \mathbb{F}$, with $(f + g)(x) = f(x) + g(x)$ and $(\alpha \cdot f)(x) = \alpha \cdot f(x)$.
3. $P(\mathbb{R})$ = vector space of polynomials over $\mathbb{R}$.

1.2.2 Subspaces

Definition 1.11 · Subspace

W is a **subspace** of V if $W \subseteq V$ with the same operations and field, which forms a vector space on its own. I.e., W is closed under the operations.

Lemma 1.8 · Subspace Test

A subset W of a vector space V forms a subspace iff $\forall \alpha, \beta \in \mathbb{F}, \forall v, w \in W$: $\alpha v + \beta w \in W$.

Example: Is $W = \{(x, y) : 2x - y + 3 = 0\}$ a subspace? **No**, because W doesn't contain $\vec{0}$.
Is $W = \{(x, y, z) : 5x - 2y + 5z = 0\}$ a subspace? **Yes**, because for any two solutions, their linear combination is also a solution.

1.2.3 Span and Linear Independence

Definition 1.12 · Span

The **span** of vectors $v_1, \dots, v_n \in V$ is

$$\operatorname{span}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^{n} \alpha_i v_i : \alpha_1, \dots, \alpha_n \in \mathbb{F} \right\}$$

This is the set of all linear combinations.

Lemma 1.9 · Span is a Subspace

Any set of vectors $v_1, \dots, v_n$: span$(v_1, \dots, v_n)$ is a subspace of V.

Definition 1.13 · Linear Independence

Vectors $v_1, \dots, v_n \in V$ are **linearly independent** if the only linear combination $c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = \vec{0}$ is the "trivial" one where $c_1 = \cdots = c_n = 0$.
Conversely, we call $v_1, \dots, v_n$ **linearly dependent** if $\exists c_1, \dots, c_n \in \mathbb{F}$, at least one nonzero, s.t. $c_1 v_1 + \cdots + c_n v_n = \vec{0}$.

Lemma 1.10 · Dependence Means Redundancy

$v_1, \dots, v_n \in V$ are linearly dependent iff $\exists k \in \{1, \dots, n\}$ such that $v_k = c_1 v_1 + \cdots + c_{k-1} v_{k-1}$ for some $c_1, \dots, c_{k-1} \in \mathbb{F}$.

Proof. ($\Leftarrow$): Rearrange $v_k = c_1 v_1 + \cdots + c_{k-1} v_{k-1}$ to get $c_1 v_1 + \cdots + c_{k-1} v_{k-1} + (-1) v_k + 0 \cdot v_{k+1} + \cdots + 0 \cdot v_n = \vec{0}$.
($\Rightarrow$): $c_1 v_1 + \cdots + c_n v_n = \vec{0}$ with not all zero. Consider the maximum k s.t. $c_k \neq 0$. Then $c_k v_k = -c_1 v_1 - \cdots - c_{k-1} v_{k-1}$, so $v_k = -\frac{c_1}{c_k} v_1 - \cdots - \frac{c_{k-1}}{c_k} v_{k-1}$. ■

Exercises

1. **(1.1)** Using only the field axioms (F1–F6), prove: (i) $a > 0 \Rightarrow -a < 0$, (ii) $a > 0 \Rightarrow \frac{1}{a} > 0$, (iii) $a > b > 0 \Rightarrow \frac{1}{a} < \frac{1}{b}$.

2. **(1.2)** If $S \subset \mathbb{R}$ is nonempty and bounded above, prove there exists a sequence $\{a_n\} \subset S$ with $\lim a_n = \sup S$.

3. **(1.3)** Prove that every positive real number has a positive square root. *Hint: Consider* $S = \{x : x^2 < a, x > 0\}$.

4. **(1.4)** Let $\mathbb{Z}_2 = \{0, 1\}$ with $1 + 1 = 0$. Show that $\mathbb{Z}_2$ is a field but cannot be ordered.

5. **(1.5)** Use the Archimedean property to prove that $\lim \frac{1}{n} = 0$.

6. **(1.6)** Decide whether each is a subspace of $\mathbb{R}^3$: (a) $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$, (b) $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$.

7. **(1.7)** If v_1, v_2, v_3, v_4 are linearly independent, are $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4 - v_1$ linearly independent?

8. **(1.8)** Prove that for subspaces $U_1, U_2 \subset V$, $U_1 \cup U_2$ is a subspace iff $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$.

Week 2

Subsequences and Continuity

2.1 Analysis: Mon/Tue

2.1.1 Subsequences and Bolzano-Weierstrass

Theorem 2.1 · Convergent $\implies$ bounded

If $\{a_n\}_{n=1}^{\infty} \subseteq \mathbb{R}$ is a convergent sequence then $\{a_n\}_{n=1}^{\infty}$ is bounded.

Proof. We WTS there exists $c > 0$ s.t. $|a_n| < c$ for all $n \geq N$. Using the definition of convergence with $\varepsilon = 1$, we know that there exists $L \in \mathbb{R}$ and $N \in \mathbb{N}$ s.t. $|a_n - L| < 1$ for all $n > N$. Hence, $a_n \leq \max\{|L| + 1, |a_1|, ..., |a_N|\} = c$. ■

Remark. The concept of the subsequence is very important in real analysis. Before we get to them, consider the sequence $-1, 1, -1, 1, -1, ...$ that is bounded but not convergent. Observe that if we remove a lot of terms, the resulting sequence is convergent. (It becomes $1, 1, 1, ...$ or $-1, -1, -1, ...$)

Definition 2.1 · Subsequence

If $\{a_n\}_{n=1}^{\infty}$ is a subsequence, $\{a_{f(k)}\}_{k=1}^{\infty}$ if a subsequence of a_n if $f : \mathbb{N} \to \mathbb{N}$ is strictly increasing ($f(k) < f(k+1)$).

Remark. Common notation for a subsequence is $\{a_{n(k)}\}_{n=1}^{\infty}$ or $\{a_{n_k}\}_{k=1}^{\infty}$

Theorem 2.2 · Bolzano-Weierstrass

Any bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Sketch. Divide the bounding interval $[c, d]$ in half repeatedly. Choose the half containing infinitely many terms. This creates nested intervals converging to a point L. For an in-depth proof of Bolzano-Weierstrass, visit https://personal.math.ubc.ca/~feldman/m320/bolzano.pdf ■

2.1.2 Continuity

Definition 2.2 · Continuity

$f : [a, b] \to \mathbb{R}$ is **continuous** at c if:

$$\lim_{x \to c} f(x) = f(c)$$

Equivalently (Sequential Criterion), for any sequence $x_n \to c$, $f(x_n) \to f(c)$.

Theorem 2.3 · Extreme Value Theorem (EVT)

If f is continuous on a closed interval $[a, b]$, then f is bounded and achieves its maximum and minimum.

Theorem 2.4 · Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and A lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = A$.

2.2 Linear Algebra: Thu/Fri

2.2.1 Basis and Dimension

Lemma 2.5 · Replacement Lemma

If $W = span(v_1, \ldots, v_n)$ and $w_1, \ldots, w_k \in W$ are linearly independent, then we can replace k of the v_i's with the w_i's while preserving $span(\cdots) = W$.

Sketch. We start with $(v_1, \ldots, v_n)$ and replace them one by one with $w_1, \ldots, w_k$, preserving that $\text{span}(\cdots) = W$.
Since $w_1 \in W$, we have $w_1 = c_1 v_1 + \cdots + c_n v_n$. There is some v_j which is a linear combination of the others (including w_1), so we can replace v_j with w_1 and keep the same span. Repeat. ■

Lemma 2.6 · Bound on Independent Vectors

If $V = span(v_1, \ldots, v_n)$ and $w_1, \ldots, w_k \in V$ are linearly independent, then $k \leq n$.

Proof. Assume not. By the replacement lemma, $V = \text{span}(w_1, \ldots, w_n)$. Then $w_{n+1} \in V = \text{span}(w_1, \ldots, w_n)$, so $\{w_1, \ldots, w_n, w_{n+1}\}$ is linearly dependent, contradicting independence of $w_1, \ldots, w_k$. ■

Definition 2.3 · Basis

$\{v_1, \ldots, v_n\} \subseteq V$ is called a **basis** of V if:

(B1) $\operatorname{span}(v_1, \ldots, v_n) = V$

(B2) $\{v_1, \ldots, v_n\}$ is linearly independent

Example: $\{e_1, e_2, e_3\}$ is a basis for $\mathbb{R}^3$, where $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$.

Lemma 2.7 · Unique Coordinates

*If $\{v_1, \ldots, v_n\}$ is a basis, then any $v \in V$ can be written **uniquely** as $v = \sum_{i=1}^{n} \alpha_i v_i$.*

Proof. That any v can be written this way follows from (B1). For uniqueness, suppose $v = \sum \alpha_i v_i = \sum \beta_i v_i$. Then $\sum(\alpha_i - \beta_i)v_i = \vec{0}$. By (B2), $\alpha_i - \beta_i = 0$ for all i, so $\alpha_i = \beta_i$. ■

Definition 2.4 · Dimension

Any two bases of a vector space V have the same number of elements. We call this number the **dimension** of V, written $\dim V$.

Proof that dimension is well-defined. Suppose $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ are both bases. Whichever is larger would give linearly independent vectors in the span of the smaller, contradicting the bound lemma. ■

2.2.2 Inner Products

Definition 2.5 · Dot Product

For $v, w \in \mathbb{R}^n$ with $v = (v_1, \ldots, v_n)$ and $w = (w_1, \ldots, w_n)$, define

$$v \cdot w = \sum_{i=1}^{n} v_i w_i$$

We say v is **orthogonal** to w (write $v \perp w$) if $v \cdot w = 0$.

Theorem 2.8 · Properties of Dot Product

1. $v \cdot w = w \cdot v$
2. $(\alpha v + \beta w) \cdot u = \alpha(v \cdot u) + \beta(w \cdot u)$
3. $v \cdot v \geq 0$, *with equality iff* $v = \vec{0}$

Definition 2.6 · Norm

Given $v \in \mathbb{R}^n$, define $\|v\| = \sqrt{v \cdot v}$.

Theorem 2.9 · Pythagorean Theorem

If $v \cdot w = 0$, *then* $\|v\|^2 + \|w\|^2 = \|v+w\|^2$.

Proof. $\|v+w\|^2 = (v+w)\cdot(v+w) = v\cdot v + 2(v\cdot w) + w\cdot w = \|v\|^2 + \|w\|^2$. ∎

Theorem 2.10 · Cauchy-Schwarz Inequality

$|v \cdot w| \leq \|v\|\|w\|$ *for all* $v, w \in \mathbb{R}^n$.

Proof. If $w = 0$, both sides are 0. Assume $w \neq 0$.
Decompose $v = v_\parallel + v_\perp$ where $v_\parallel = \alpha w$ for some α and $v_\perp \cdot w = 0$.
From $v \cdot w = (v_\parallel + v_\perp)\cdot w = \alpha w \cdot w = \alpha\|w\|^2$, we get $\alpha = \frac{v\cdot w}{\|w\|^2}$.
By Pythagorean theorem: $\|v\|^2 = \|v_\parallel\|^2 + \|v_\perp\|^2 \geq \|v_\parallel\|^2 = \frac{(v\cdot w)^2}{\|w\|^2}$.
Hence $(v\cdot w)^2 \leq \|v\|^2\|w\|^2$. ∎

Theorem 2.11 · Triangle Inequality

$\|v+w\| \leq \|v\| + \|w\|$.

Exercises

1. **(2.1)** (*Sandwich Theorem*) If $\lim a_n = \lim b_n = L$ and $a_n \leq c_n \leq b_n$ for all n, prove $\lim c_n = L$.
2. **(2.2)** (*Heine-Borel*) Let $\mathcal{F}$ be a collection of open intervals covering $[a,b]$. Prove there is a finite subcollection that still covers $[a,b]$.
3. **(2.3)** If $\lim f(x_n) = f(c)$ for every sequence $\{x_n\} \to c$, prove f is continuous at c.
4. **(2.4)** Let $f(x) = 0$ if x is irrational, and $f(p/q) = 1/q$ for p/q in lowest terms. Prove f is continuous at every irrational.
5. **(2.5)** Let V be finite-dimensional and $U \subseteq V$ a subspace. Prove $U = V$ iff $\dim U = \dim V$.
6. **(2.6)** If $\dim U + \dim W > \dim V$ for subspaces $U, W \subseteq V$, prove $U \cap W \neq \{0\}$.
7. **(2.7)** Find a basis for $V = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = 0\}$ and extend it to a basis of $\mathcal{P}_4(\mathbb{R})$.
8. **(2.8)** If $\{v_1, \ldots, v_m\}$ is linearly independent and $w \in V$, prove $\dim(\operatorname{span}\{v_1 + w, \ldots, v_m + w\}) \geq m - 1$.

Week 3

Differentiation and Topology

3.1 Analysis: Mon/Tue

3.1.1 Differentiable Functions in $\mathbb{R}$

Definition 3.1 · Differentiability

Given $f : [a, b] \to \mathbb{R}$, $c \in [a, b]$, we say f is **differentiable** at c if

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

exists. We denote the limit, if it exists, by $f'(c)$ or $\frac{df}{dx}(c)$.

Remark. Some basic derivative laws:

1. If f, g are differentiable at c, then $\alpha f + \beta g$ is differentiable at c with $(\alpha f + \beta g)'(c) = \alpha f'(c) + \beta g'(c)$.
2. If $f : [a, b] \to \mathbb{R}$ is constant, then $f'(c) = 0$ for all $c \in (a, b)$.
3. If $f(x) = x$, then $f'(c) = 1$ for all $c \in (a, b)$.

Theorem 3.1 · Mean Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem 3.2 · Rolle's Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Suppose $f(a) = 0 = f(b)$. Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. We first prove Rolle's theorem and use it to prove MVT.
If f is constant, then $f'(c) = 0$ and we are done. Otherwise, either $\exists x_0 \in (a, b)$ with $f(x_0) > 0$ or $\exists x_0 \in (a, b)$ with $f(x_0) < 0$.
In the former case, there exists $c \in [a, b]$ such that $f(x) \leq f(c)$ for all $x \in [a, b]$. Since $f(c) \geq f(x_0) > 0$, we know $c \in (a, b)$.
Claim: $f'(c) = 0$.
We know $f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$.
Take $\{x_n\}$ with $\lim x_n = c$ and $c < x_n < b$. Then

$$f'(c) = \lim_{n \to \infty} \frac{f(x_n) - f(c)}{x_n - c} \leq 0$$

since the numerator is ≤ 0 and denominator is > 0.
On the other hand, take $\{x_n\}$ with $\lim x_n = c$ and $a < x_n < c$. Then

$$f'(c) = \lim_{n \to \infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0$$

since both numerator and denominator are ≤ 0.
Hence $f'(c) = 0$.
MVT from Rolle's: Given f, define $g : [a, b] \to \mathbb{R}$ by

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Note that $g(a) = 0$ and $g(b) = 0$. By Rolle's theorem, there exists $c \in (a, b)$ such that

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

which gives $f'(c) = \frac{f(b) - f(a)}{b - a}$. ■

3.1.2 Sequences in $\mathbb{R}^n$

Definition 3.2 · Sequence in $\mathbb{R}^n$

A sequence in $\mathbb{R}^n$ is a map $\mathbb{N} \to \mathbb{R}^n$. We denote by $\{a^{(k)}\}_{k=1}^{\infty}$, where $a^{(k)} = (a_1^{(k)}, \ldots, a_n^{(k)})$.

Definition 3.3 · Limit in $\mathbb{R}^n$

Given $\{a^{(k)}\}_{k=1}^{\infty} \subseteq \mathbb{R}^n$ and $a \in \mathbb{R}^n$, we say $\lim_{k \to \infty} a^{(k)} = a$ if for all $\varepsilon > 0$, there exists $\mathcal{K} \in \mathbb{N}$ such that $\|a^{(k)} - a\| < \varepsilon$ whenever $k \geq \mathcal{K}$.
Here $\|b\| = \sqrt{b \cdot b} = \sqrt{\sum_{i=1}^{n} b_i^2}$.

Lemma 3.3 · Component-wise Convergence

$\{a^{(k)}\}_{k=1}^{\infty} \subseteq \mathbb{R}^n$, $a \in \mathbb{R}^n$. Then $\lim a^{(k)} = a$ if and only if $\lim a_i^{(k)} = a_i$ for all $i = 1, \ldots, n$.

Proof. ($\Leftarrow$): Let $\varepsilon > 0$. Since $\lim a_i^{(k)} = a_i$ for every i, there exists $\mathcal{K}_i \in \mathbb{N}$ such that $|a_i^{(k)} - a_i| < \frac{\varepsilon}{\sqrt{n}}$ whenever $k \geq \mathcal{K}_i$.
Let $\mathcal{K} = \max\{\mathcal{K}_1, \dots, \mathcal{K}_n\}$. Then

$$\|a^{(k)} - a\| = \left(\sum_{i=1}^{n} |a_i^{(k)} - a_i|^2\right)^{1/2} < \left(\sum_{i=1}^{n} \frac{\varepsilon^2}{n}\right)^{1/2} = \varepsilon$$

($\Rightarrow$): Exercise. ■

Definition 3.4 · Bounded Set

$S \subseteq \mathbb{R}^n$ is **bounded** if there exists $c > 0$ such that $\|x\| < c$ for all $x \in S$.

Theorem 3.4 · Bolzano-Weierstrass in $\mathbb{R}^n$

Any bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Sketch. Prove by induction on n. The base case was proven in Week 1. Assume the result up to dimension $n-1$.
Let $\{a^{(k)}\}_{k=1}^{\infty} \subseteq \mathbb{R}^n$. Write $a^{(k)} = (a^{1(k)}, a_n^{(k)})$ where $a^{1(k)} = (a_1^{(k)}, \dots, a_{n-1}^{(k)})$.
By induction, there exists a subsequence $\{a^{(k_j)}\}$ such that $\lim_{j\to\infty} a^{1(k_j)}$ exists.
By BW in $\mathbb{R}$, there exists a further subsequence $\{a_n^{(k_{j_\ell})}\}$ such that $\lim_{\ell\to\infty} a_n^{(k_{j_\ell})}$ exists.
Then $\{a^{(k_{j_\ell})}\}_{\ell=1}^{\infty} \subseteq \mathbb{R}^n$ is convergent. ■

3.1.3 Open and Closed Sets

Definition 3.5 · Ball

For $x \in \mathbb{R}^n$ and $\rho > 0$, define $B_\rho(x) = \{y \in \mathbb{R}^n : \|x - y\| < \rho\}$.

Definition 3.6 · Open Set

$U \subseteq \mathbb{R}^n$ is **open** if for all $x \in U$, there exists $\rho > 0$ such that $B_\rho(x) \subseteq U$.

Definition 3.7 · Limit Point

$A \subseteq \mathbb{R}^n$. We say $b \in \mathbb{R}^n$ is a **limit point** of A if there exists $\{a^{(k)}\}_{k=1}^{\infty}$ with $a^{(k)} \in A$ such that $b = \lim_{k\to\infty} a^{(k)}$.

Definition 3.8 · Closed Set

$K \subseteq \mathbb{R}^n$ is **closed** if it contains all its limit points.

Remark. A set can be both open and closed (e.g., $\mathbb{R}^n$ and $\varnothing$), or neither (e.g., $(a, b]$).

Theorem 3.5 · Union of Open Sets

If $\{U_\alpha\}_{\alpha\in\Gamma}$ is a collection of open sets in $\mathbb{R}^n$, then $\bigcup_{\alpha\in\Gamma} U_\alpha$ is open.
If $\{K_\alpha\}_{\alpha\in\Gamma}$ is a collection of closed sets, then $\bigcap_{\alpha\in\Gamma} K_\alpha$ is closed.

Proof. Let $x \in \bigcup_{\alpha\in\Gamma} U_\alpha$. Then there exists $\alpha_0 \in \Gamma$ such that $x \in U_{\alpha_0}$.
Since U_{α_0} is open, there exists $\rho > 0$ such that $B_\rho(x) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha\in\Gamma} U_\alpha$. ∎

Theorem 3.6 · Open-Closed Duality

$U \subseteq \mathbb{R}^n$ is open if and only if $\mathbb{R}^n \setminus U$ is closed.

Proof. $(\Rightarrow)$: Let U be open and let $z \in \mathbb{R}^n$ be a limit point of $\mathbb{R}^n \setminus U$. We want to show $z \notin U$. If $z \in U$, there exists $\rho > 0$ such that $B_\rho(z) \subseteq U$. But since z is a limit point of $\mathbb{R}^n \setminus U$, there exists $y \in \mathbb{R}^n \setminus U$ with $\|y - z\| < \rho$. Hence $y \in B_\rho(z) \subseteq U$ and $y \in \mathbb{R}^n \setminus U$, contradiction.
$(\Leftarrow)$: Prove the contrapositive. If U is not open, then $\exists x \in U$ such that for all $\rho > 0$, there exists $y_\rho \in B_\rho(x)$ with $y_\rho \in \mathbb{R}^n \setminus U$. Taking $\rho = 1/j$, we get a sequence converging to x, so x is a limit point of $\mathbb{R}^n \setminus U$ but $x \in U$, hence $\mathbb{R}^n \setminus U$ is not closed. ∎

Definition 3.9 · Compact Set

$K \subseteq \mathbb{R}^n$ is **compact** if for all sequences $\{a^{(k)}\}_{k=1}^\infty \subseteq K$, there exists a subsequence $\{a^{(k_j)}\}$ and $a \in K$ such that $\lim_{j\to\infty} a^{(k_j)} = a$.

Remark. In $\mathbb{R}^n$: closed + bounded $\Leftrightarrow$ compact.

3.2 Linear Algebra: Thu/Fri

3.2.1 Linear Maps

Definition 3.10 · Linear Map

A function $T : V \to W$ (over field $\mathbb{F}$) is a **linear map** if

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$$

for all $\alpha, \beta \in \mathbb{F}$ and $x, y \in V$.

Remark. $T(0) = 0$ always!

Example:

1. $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(x_1, x_2) = (x_1, x_1x_2)$ is NOT linear.
2. $T(x_1, x_2) = (x_1, \lambda x_2)$ for fixed $\lambda \in \mathbb{R}$ IS linear.

3. Rotation: $(x_1, x_2) \mapsto (x_1 \cos\theta - x_2 \sin\theta, x_1 \sin\theta + x_2 \cos\theta)$ is linear.

3.2.2 Matrix Representation

Fix a basis $\{v_1, \ldots, v_n\}$ of V and $\{w_1, \ldots, w_m\}$ of W.

How does T act on the basis vectors? Write

$$T(v_j) = a_{1j}w_1 + a_{2j}w_2 + \cdots + a_{mj}w_m$$

For any vector $x \in V$ with $x = \xi_1 v_1 + \cdots + \xi_n v_n$:

$$\begin{aligned} T(x) &= \xi_1 T(v_1) + \cdots + \xi_n T(v_n) \\ &= (a_{11}\xi_1 + \cdots + a_{1n}\xi_n)w_1 + \cdots + (a_{m1}\xi_1 + \cdots + a_{mn}\xi_n)w_m \end{aligned}$$

Definition 3.11 · Matrix of a Linear Map

Given a linear map $T : V \to W$ with $T(v_j) = \sum_i a_{ij}w_i$, the **matrix of** T under these bases is

$$A = (a_{ij}) \in \mathbb{F}^{m \times n}$$

with n columns ($= \dim V$) and m rows ($= \dim W$).

Definition 3.12 · Linear Operator

A linear map from V to V is called a **linear operator** on V. Its matrix is square ($n \times n$).

3.2.3 Composition and Matrix Multiplication

Given $T_1 : V_1 \to V_2$ and $T_2 : V_2 \to V_3$, define $T_2 \circ T_1 : V_1 \to V_3$ by $(T_2 \circ T_1)(x) = T_2(T_1(x))$.

If A is the matrix for T_1 and B is the matrix for T_2, then the matrix for $T_2 \circ T_1$ is $C = B \cdot A$ where

$$c_{ik} = \sum_j b_{ij}a_{jk} = (i\text{th row of } B) \cdot (k\text{th column of } A)$$

Remark. Matrix multiplication is associative: $A(BC) = (AB)C$, but NOT commutative: $AB \neq BA$ in general.

Exercises

1. **(3.1)** Prove that a continuous function $f : [a, b] \to \mathbb{R}$ is uniformly continuous.
2. **(3.2)** Show that $f(x) = 1/x$ on $(0, 1)$ is *not* uniformly continuous.
3. **(3.3)** (*Cauchy criterion*) Prove that $\{a_n\}$ converges iff $\forall \varepsilon > 0, \exists N$ s.t. $|a_m - a_n| < \varepsilon$ for $m, n \geq N$.
4. **(3.4)** Prove: $E \subseteq A$ is relatively open in A iff $\exists$ open $U \subseteq \mathbb{R}^n$ with $E = A \cap U$.

5. **(3.5)** Prove the identity $\frac{1}{2}\sum_{i,j}(a_i b_j - a_j b_i)^2 = \|a\|^2\|b\|^2 - (a \cdot b)^2$ and use it to prove Cauchy-Schwarz.
6. **(3.6)** Prove that equality holds in the triangle inequality iff $y = \lambda x$ for some $\lambda \geq 0$ (or one is zero).
7. **(3.7)** Prove $(AB)^T = B^T A^T$ directly from the definition of matrix multiplication.
8. **(3.8)** Find the matrix of rotation by angle θ in $\mathbb{R}^2$. What is the matrix for reflection across $y = x$?

Week 4

Continuity and Metric Spaces

4.1 Linear Algebra: Thu/Fri (continued from Week 3)

4.1.1 Kernel and Range

Definition 4.1 · Kernel and Range

For a linear map $T : V \to W$:

1. $\ker(T) = \{v \in V : T(v) = 0\}$
2. $\text{Range}(T) = \{y \in W : \exists v \in V, T(v) = y\}$

Lemma 4.1 · Kernel and Range are Subspaces

$\ker(T)$ and Range(T) are subspaces of V and W, respectively.

Proof. Take $u, v \in \ker(T)$ and $\alpha, \beta \in \mathbb{F}$:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v) = 0 + 0 = 0$$

Take $x, y \in \text{Range}(T)$ with $x = T(u)$, $y = T(v)$:

$$\alpha x + \beta y = \alpha T(u) + \beta T(v) = T(\alpha u + \beta v) \in \text{Range}(T)$$

■

Remark. For a matrix $A \in \mathbb{F}^{m \times n}$:

- $\text{Range}(A) = \text{span}(\text{columns of } A) = \text{column space of } A$
- $\ker(A) = \{x \in \mathbb{F}^n : Ax = 0\}$
- Row space: $R(A) = \text{span}(\text{rows of } A)$

Definition 4.2 · Injective, Surjective, Bijective

- f is **injective**: if $u \neq v$, then $f(u) \neq f(v)$
- f is **surjective**: $\forall y \in W$, $\exists u \in V$ s.t. $f(u) = y$
- f is **bijective**: both injective and surjective

Remark. For linear maps:

- T is surjective $\Leftrightarrow$ $\text{Range}(T) = W$
- T is injective $\Leftrightarrow$ $\ker(T) = \{0\}$

4.1.2 Rank-Nullity Theorem

Theorem 4.2 · Rank-Nullity Theorem

If $T : V \to W$ is linear and V is finite dimensional, then

$$\dim(\ker(T)) + \dim(\mathit{Range}(T)) = \dim V$$

Proof. Construct a basis $\{u_1, \ldots, u_k\}$ of $\ker(T)$ (assuming $\dim(\ker(T)) = k$; if $k = 0$, this is empty).
Extend to a basis of V: $\{u_1, \ldots, u_k, v_{k+1}, \ldots, v_n\}$.
Claim: $\text{Range}(T) = \text{span}(T(v_{k+1}), \ldots, T(v_n))$.
If $y \in \text{Range}(T)$, then $y = T(x)$ for some $x = \alpha_1 u_1 + \cdots + \alpha_k u_k + \alpha_{k+1} v_{k+1} + \cdots + \alpha_n v_n$, so $y = \alpha_{k+1} T(v_{k+1}) + \cdots + \alpha_n T(v_n)$.
Claim: $T(v_{k+1}), \ldots, T(v_n)$ are linearly independent.
Assume $\beta_{k+1} T(v_{k+1}) + \cdots + \beta_n T(v_n) = 0$. Then $T(\beta_{k+1} v_{k+1} + \cdots + \beta_n v_n) = 0$, so $\beta_{k+1} v_{k+1} + \cdots + \beta_n v_n \in \ker(T) = \text{span}(u_1, \ldots, u_k)$.
By linear independence of the full basis, $\beta_{k+1} = \cdots = \beta_n = 0$.
Hence $\dim(\text{Range}(T)) = n - k$. ■

Definition 4.3 · Rank

Define the **rank** of A as $\text{rank}(A) = \dim(\text{Range}(A)) = \dim(\text{column space})$.

Lemma 4.3 · Rank from RREF

$\dim(\ker(A)) = n - \#\mathit{pivots}$ *and* $\mathit{rank}(A) = \#\mathit{pivots}$.

4.2 Analysis: Mon/Tue

4.2.1 Continuity in Multiple Dimensions

Definition 4.4 · Continuity in $\mathbb{R}^n$

Given $A \subseteq \mathbb{R}^n$, $f : A \to \mathbb{R}^m$, $a \in A$, we say f is **continuous at** a if for all $\varepsilon > 0$, there exists $\delta > 0$ such that $\|f(x) - f(a)\| < \varepsilon$ whenever $\|x - a\| < \delta$ and $x \in A$.
f is **continuous** if it is continuous at all $a \in A$.

Remark. Some useful facts:

1. $a \in A$ is an **isolated point** if there exists $\rho > 0$ such that $B_\rho(a) \cap A = \{a\}$.
2. f is continuous at a non-isolated point a iff $\lim_{x\to a} f(x) = f(a)$.
3. Equivalently: f is continuous at a iff for any sequence $\{a^{(k)}\} \subseteq A$ with $\lim a^{(k)} = a$, we have $\lim f(a^{(k)}) = f(a)$.

Theorem 4.4 · Extreme Value Theorem (Compact Version)

If $A \subseteq \mathbb{R}^n$ is compact and $f : A \to \mathbb{R}$ is continuous, then f is bounded and achieves its maximum and minimum.

Proof. Suppose not. Then for all $k \in \mathbb{N}$, there exists $x^{(k)} \in A$ such that $|f(x^{(k)})| \geq k$.
Since A is compact, there exists a subsequence $\{x^{(k_j)}\}$ and $a \in A$ such that $\lim_{j\to\infty} x^{(k_j)} = a$.
By continuity, $\lim_{j\to\infty} f(x^{(k_j)}) = f(a)$, which is convergent and thus bounded. But $|f(x^{(k_j)})| \geq k_j \to \infty$, contradiction. ∎

4.2.2 Metric Spaces

Definition 4.5 · Metric Space

(X, d) is a **metric space** if X is a set and $d : X \times X \to \mathbb{R}$ satisfies:

(M1) $d(x, y) \geq 0$ for all $x, y \in X$, with equality iff $x = y$.

(M2) $d(x, y) = d(y, x)$ for all $x, y \in X$.

(M3) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality).

Example:

1. $X = \mathbb{R}^n$, $d(x, y) = \|x - y\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
2. $X \subseteq \mathbb{R}^n$, $d(x, y) = \|x - y\|$
3. Discrete metric: $d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$

4. Any normed vector space $(V, \|\cdot\|)$ with $d(x,y) = \|x-y\|$

Remark. In a metric space (X,d):

1. $\{x_n\} \subseteq X$ converges to $x \in X$ if $\forall \varepsilon > 0$, $\exists N$ such that $d(x_n, x) < \varepsilon$ for $n \geq N$.
2. Ball: $B_\rho(y) = \{x \in X : d(x,y) < \rho\}$
3. x is a limit point of A if $\exists \{a_n\} \subseteq A$ with $\lim a_n = x$.
4. A is open if $\forall x \in A$, $\exists \rho > 0$ with $B_\rho(x) \subseteq A$.
5. A is closed if it contains all its limit points.
6. A is bounded if $\exists y \in X$, $\rho > 0$ with $A \subseteq B_\rho(y)$.
7. A is compact if every sequence in A has a convergent subsequence with limit in A.

Theorem 4.5 · Compact implies Closed and Bounded

Let (X,d) be a metric space and $A \subseteq X$. If A is compact, then A is closed and bounded.

Proof. **Boundedness:** Suppose not. Then for all $y \in X$ and $\rho > 0$, $A \not\subseteq B_\rho(y)$. So there exists $a_n \in A$ with $d(a_n, y) \geq n$. By compactness, some subsequence $\{a_{n_k}\}$ converges to $a \in A$, so $\{a_{n_k}\}$ is bounded. But $d(a_{n_k}, y) \geq n_k \geq k \to \infty$, contradiction.
Closedness: Suppose $x \in X$ is a limit point of A. We WTS $x \in A$. By definition, there exists $\{a_n\} \subseteq A$ with $\lim a_n = x$. By compactness, there exists $\{a_{n_k}\}$ with $\lim a_{n_k} = a \in A$. Since limits are unique, $x = a \in A$. ■

Definition 4.6 · Continuity in Metric Spaces

Given metric spaces (X, d_X), (Y, d_Y) and $f : X \to Y$:

1. f is **continuous at** $c \in X$ if $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $d_Y(f(x), f(c)) < \varepsilon$ whenever $d_X(x,c) < \delta$.
2. f is **continuous** if it is continuous at all points.

Theorem 4.6 · Image of Compact is Compact

Let (X, d_X), (Y, d_Y) be metric spaces with X compact and $f : X \to Y$ continuous. Then $f(X) = \{f(x) : x \in X\}$ is compact.

Proof. Let $\{y_n\} \subseteq f(X)$. Then there exists $x_n \in X$ with $y_n = f(x_n)$.
By compactness of X, there exists $\{x_{n_k}\}$ and $x \in X$ with $\lim x_{n_k} = x$.
By continuity of f, $\lim y_{n_k} = \lim f(x_{n_k}) = f(x) \in f(X)$. ■

4.3 Linear Algebra: Thu/Fri (continued)

See Week 3 for kernel, range, and rank-nullity theorem content.

Exercises

1. **(4.1)** Let $f : [a, b] \to \mathbb{R}^n$ be differentiable. Prove $\exists c \in (a, b)$ with $\|f(b) - f(a)\| \leq (b - a)\|f'(c)\|$.
2. **(4.2)** Prove that $\bar{A}$ (closure of A) is closed in any metric space.
3. **(4.3)** Prove that $f : X \to Y$ continuous and $U \subseteq Y$ open implies $f^{-1}(U)$ is open in X.
4. **(4.4)** If A is dense in X and $f, g : X \to Y$ are continuous with $f|_A = g|_A$, prove $f = g$.
5. **(4.5)** Prove that $\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$.
6. **(4.6)** Prove that $\dim\ker(T \circ S) \leq \dim\ker T + \dim\ker S$.
7. **(4.7)** Find the matrix of orthogonal projection onto $W = \operatorname{span}\{v\}$ where v is a unit vector in $\mathbb{R}^n$.
8. **(4.8)** Solve the system with augmented matrix $\begin{pmatrix} 1 & 0 & 1 & 1 & 1 & \mid & 1 \\ 2 & 1 & 1 & 3 & 1 & \mid & 1 \\ 1 & 1 & 2 & 0 & 2 & \mid & 2 \\ 0 & 0 & 1 & -1 & 1 & \mid & 1 \end{pmatrix}$.

Week 5
Series and Differentiation in Higher Dimensions

5.1 Linear Algebra: Thu/Fri (Gaussian Elimination)

5.1.1 Solving Linear Systems

Lemma 5.1 · Solutions to $Ax = 0$

1. *Solutions to* $Ax = 0 = \ker(A)$
2. $Ax = b$ *has a solution* $\Leftrightarrow b \in Range(A)$

Moreover, if $\bar{x}$ is a particular solution, then the set of all solutions is $\{\bar{x} + w : w \in \ker(A)\} = \bar{x} + \ker(A)$.

Proof. Suppose $x = \bar{x} + w$ with $w \in \ker(A)$. Then $Ax = A(\bar{x} + w) = A\bar{x} + Aw = b + 0 = b$. Conversely, if x is a solution, then $A(x - \bar{x}) = b - b = 0$, so $x - \bar{x} \in \ker(A)$. ∎

5.1.2 Row Operations

Definition 5.1 · Row Operations

(R1) Switch two rows

(R2) Multiply row i by nonzero constant $\lambda \in \mathbb{F}$

(R3) Replace row i by (row i + $\lambda\cdot$ row j)

Lemma 5.2 · Row Operations Preserve Kernel

1. *Performing row operations on A doesn't change* $\ker(A)$ *(it may change Range(A) though!)*
2. *Performing row operations on the augmented matrix $(A \mid b)$ does not change the solution set.*

Lemma 5.3 · RREF Existence

Every (augmented) matrix can be reduced to RREF by finitely many row operations.

Example: To find $\ker(A)$: reduce to RREF. Non-pivot variables are "free." E.g., if RREF gives

$$\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -3 \end{pmatrix}$$

then $\ker(A) = \{(-3\beta, 3\beta, \beta) : \beta \in \mathbb{F}\}$.

5.1.3 Orthogonal Complements

Definition 5.2 · Orthogonal Complement

For $V \subseteq \mathbb{F}^n$ a subspace,

$$V^\perp = \{x \in \mathbb{F}^n : x \cdot v = 0 \text{ for all } v \in V\}$$

Theorem 5.4 · Dimension of Orthogonal Complement

1. $\dim V + \dim V^\perp = n$
2. $(V^\perp)^\perp = V$
3. *If* $\mathbb{F} = \mathbb{R}$, *then* $V \cap V^\perp = \{0\}$
4. *If* $\mathbb{F} = \mathbb{R}$, *then* $\forall x \in \mathbb{R}^n$, $\exists!$ *decomposition* $x = v + v^\perp$ *with* $v \in V$, $v^\perp \in V^\perp$

Proof of (1). Let $\{v_1, \ldots, v_k\}$ be a basis of V. Form matrix A with rows v_i. Note $\ker(A) = V^\perp$ since $Ax = 0$ iff $v_i \cdot x = 0$ for all i.
Hence $\dim(\text{Range}(A)) + \dim(V^\perp) = n$, and $\dim(\text{Range}(A)) = k = \dim V$. ∎

5.2 Analysis: Mon/Tue

5.2.1 Series

Definition 5.3 · Series

Let $\{a_k\}_{k=1}^\infty \subseteq \mathbb{R}^n$. We say $\sum_{k=1}^\infty a_k = s \in \mathbb{R}^n$ if the sequence of partial sums $S_j = \sum_{k=1}^j a_k$ satisfies $\lim_{j\to\infty} S_j = s$.
We say $\sum a_k$ **converges** if such s exists, and **diverges** otherwise.

Theorem 5.5 · Terms Must Vanish

Suppose $\{a_k\}_{k=1}^\infty \subseteq \mathbb{R}^n$ *and* $\sum a_k$ *converges. Then* $\lim_{k\to\infty} a_k = 0$.

Proof. Let $\varepsilon > 0$. We WTS $\|a_k\| < \varepsilon$ for $k \geq \mathcal{K}$.
By assumption, there exists $S \in \mathbb{R}^n$ with $\sum a_k = S$. So $\|S_j - S\| < \frac{\varepsilon}{2}$ whenever $j \geq \mathcal{J}$.
Let $\mathcal{K} = \mathcal{J} + 1$. Then for $k \geq \mathcal{K}$:

$$\|a_k\| = \|S_k - S_{k-1}\| \leq \|S - S_k\| + \|S - S_{k-1}\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

■

Theorem 5.6 · Non-negative Series

Suppose $\{a_k\}_{k=1}^{\infty} \subseteq \mathbb{R}$ with $a_k \geq 0$ for all k. Then $\sum_{k=1}^{\infty} a_k$ converges iff $\{S_j\}$ is bounded above.

Proof. Observe that $\{S_j\}$ is increasing since $a_k \geq 0$. Recall that an increasing sequence converges iff it is bounded above. ■

Example: **Geometric Series:** $\sum_{k=0}^{\infty} r^k$ converges if $r \in (0, 1)$ and diverges if $r \geq 1$.

Proof. For $r \neq 1$: $S_j = 1 + r + \cdots + r^j$ and $rS_j = r + \cdots + r^{j+1}$.
So $(r-1)S_j = r^{j+1} - 1$, giving $S_j = \frac{r^{j+1}-1}{r-1}$.
This is bounded (converges to $\frac{1}{1-r}$) if $r \in (0, 1)$ and unbounded if $r > 1$. ■

Example: **Harmonic Series:** $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

Proof.

$$\begin{aligned} S_{2^j} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots \\ &\geq 1 + \frac{1}{2} + \frac{2}{4} + \frac{4}{8} + \cdots = 1 + \frac{j}{2} \to \infty \end{aligned}$$

■

Example: **p-series:** $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if $p \geq 2$ (in fact, if $p > 1$).

Proof.

$$\sum_{k=1}^{2^j-1} \frac{1}{k^p} \leq 1 + \left(\frac{1}{2^p} + \frac{1}{2^p}\right) + \left(\frac{4}{4^p}\right) + \cdots \leq 1 + \left(\frac{1}{2}\right)^{p-1} + \left(\frac{1}{2}\right)^{2(p-1)} + \cdots$$

Since $p > 1$, this is a convergent geometric series. ■

Definition 5.4 · Absolute Convergence

$\sum_{k=1}^{\infty} a_k$ is **absolutely convergent** if $\sum_{k=1}^{\infty} |a_k|$ converges.

Theorem 5.7 · Absolute Convergence Implies Convergence

If $\sum a_k$ is absolutely convergent, then it is convergent.

Proof. Let $p_k = \max\{a_k, 0\}$ and $q_k = \max\{-a_k, 0\}$. Then $|a_k| = p_k + q_k$ and $a_k = p_k - q_k$.

Since $\sum |a_k|$ converges, $\sum_{k=1}^{j} |a_k|$ is bounded. Then $\sum_{k=1}^{j} p_k \le \sum_{k=1}^{j} |a_k|$ is bounded, so $\sum p_k$ converges. Similarly $\sum q_k$ converges.
Hence $\sum a_k = \sum (p_k - q_k)$ converges. ■

Definition 5.5 · Rearrangement

Given $\{a_k\}_{k=1}^{\infty} \subseteq \mathbb{R}$, we say $\sum_{k=1}^{\infty} a_{j(k)}$ is a **rearrangement** if $j : \mathbb{N} \to \mathbb{N}$ is a bijection.

Theorem 5.8 · Rearrangements of Absolutely Convergent Series

If $\sum a_k$ converges absolutely, then any rearrangement $\sum a_{j(k)}$ also converges to the same sum.

Proof. First consider the case $a_k \ge 0$. Suppose $\sum a_k = S$.
For every $\ell \in \mathbb{N}$, define $p(\ell) = \max\{j(1), \dots, j(\ell)\}$.
Then $\sum_{k=1}^{\ell} a_{j(k)} \le \sum_{k=1}^{p(\ell)} a_k \le S$.
So $\sum a_{j(k)}$ converges. Since the partial sums are increasing,

$$\sum_{k=1}^{\infty} a_{j(k)} = \sup_{\ell} \sum_{k=1}^{\ell} a_{j(k)} \le S$$

By symmetry (the original is a rearrangement of the rearrangement), we get equality.
For the general case, use $a_k = p_k - q_k$ where both $\sum p_k$ and $\sum q_k$ converge by the previous theorem. ■

5.2.2 Differentiation in Higher Dimensions

Definition 5.6 · Derivative Matrix

Given $f : U \to \mathbb{R}^m$ where $U \subseteq \mathbb{R}^n$ is open, and $c \in U$. We say f is **differentiable at** c if there exists $A \in \mathbb{R}^{m \times n}$ such that

$$\lim_{h \to 0} \frac{\|f(c+h) - f(c) - Ah\|}{\|h\|} = 0$$

If such A exists, it is called the **derivative matrix** of f at c, denoted $Df(c)$.

Lemma 5.9 · Matrix Norm Inequality

If $\|A\| = \left(\sum_{i,j} a_{ij}^2\right)^{1/2}$, then $\|Ax\| \le \|A\|\|x\|$.

Proof. Write $A = (a_{ij})$. Then

$$\|Ax\|^2 = \sum_{i=1}^{m} (a_i \cdot x)^2 \le \sum_{i=1}^{m} \|a_i\|^2 \|x\|^2 = \|A\|^2 \|x\|^2$$

by Cauchy-Schwarz. Take square roots. ■

Theorem 5.10 · Differentiable Implies Continuous

If $f : U \to \mathbb{R}^m$ is differentiable at c, then f is continuous at c.

Proof. Since U is open, there exists $\rho > 0$ with $B_\rho(c) \subseteq U$.
Using the definition with $\varepsilon = 1$, there exists $\delta_0 \in (0, \rho)$ such that

$$\frac{\|f(c+h) - f(c) - Ah\|}{\|h\|} < 1 \text{ whenever } 0 < \|h\| < \delta_0$$

Let $\varepsilon > 0$. We WTS there exists δ such that $\|f(c+h) - f(c)\| < \varepsilon$ whenever $\|h\| < \delta$.
Let $\delta = \min\{\delta_0, \frac{\varepsilon}{2}, \frac{\varepsilon}{2\|A\|}\}$ (assuming $A \neq 0$).
Then:

$$\|f(c+h) - f(c)\| \leq \|f(c+h) - f(c) - Ah\| + \|Ah\| < \|h\| + \|A\|\|h\| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

■

5.3 Linear Algebra: Thu/Fri

Content from Linear Algebra PDF needed

Week 6

Directional Derivatives and the Chain Rule

6.1 Linear Algebra: Thu/Fri (Determinants)

6.1.1 Permutations and Signs

Definition 6.1 · Permutation

A **permutation** $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$ is a bijection. The set of all permutations on n elements is denoted S_n.

Definition 6.2 · Transposition

A **transposition** τ_{kl} is a permutation that swaps k and l and fixes everything else.

Lemma 6.1 · Permutations as Transpositions

Every permutation can be written as a composition of adjacent transpositions $\tau_{k,k+1}$.

Definition 6.3 · Inversions and Sign

For $\sigma \in S_n$, the **number of inversions** is

$$N(\sigma) = |\{(i, j) : i < j \text{ but } \sigma(i) > \sigma(j)\}|$$

Define $\text{sgn}(\sigma) = (-1)^{N(\sigma)}$.

Remark. $\text{sgn}(\tau_{kl}) = -1$ for any transposition. Each adjacent swap changes the sign.

6.1.2 The Determinant

Definition 6.4 · Determinant

For $A = (a_{ij}) \in \mathbb{F}^{n\times n}$:

$$\det(A) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\cdot a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n,\sigma(n)}$$

Remark. The meaning of the determinant is a "signed volume" — the sign captures the orientation of the vectors.

Theorem 6.2 · Properties of Determinant

1. $\det(A^T) = \det(A)$
2. *Switching two rows/columns multiplies* det *by* -1
3. det *is linear in each row/column*
4. *If two rows/columns are equal,* $\det = 0$
5. $\det(AB) = \det(A)\cdot\det(B)$

6.1.3 Inverses

Definition 6.5 · Inverse Matrix

$B \in \mathbb{F}^{n\times n}$ is an **inverse** of A if $AB = I = BA$.

Lemma 6.3 · Uniqueness of Inverse

An inverse, if it exists, is unique.

Proof. Suppose $B, C = A^{-1}$. Then $B = BI = B(AC) = (BA)C = IC = C$. ∎

Lemma 6.4 · One-Sided Implies Two-Sided

If $A, B \in \mathbb{F}^{n\times n}$ and $AB = I$, then $BA = I$.

Proof. $AB = I$ implies $\operatorname{Range}(AB) = \mathbb{F}^n$, so B is surjective. By rank-nullity, $\ker(B) = \{0\}$. Now $AB = I$ implies $ABy = y$ for all y. Hence $BABy = By$ for all y. Since B is surjective, $BAx = x$ for all x, so $BA = I$. ∎

Theorem 6.5 · Invertibility Criterion

A is invertible iff $\det(A) \neq 0$.

Proof. ($\Rightarrow$): $AA^{-1} = I$ implies $\det(A)\det(A^{-1}) = 1$, so $\det(A) \neq 0$.
($\Leftarrow$): See adjugate formula: $A \cdot \operatorname{adj}(A) = \det(A) \cdot I$. ■

Theorem 6.6 · Equivalent Statements for Invertibility

The following are equivalent:

1. A *is invertible*
2. $\exists B$ *s.t.* $AB = I$
3. $\exists B$ *s.t.* $BA = I$
4. $\det(A) \neq 0$
5. $\mathit{rref}(A) = I$
6. $\ker(A) = \{0\}$
7. $\mathit{Range}(A) = \mathbb{F}^n$
8. $\mathit{rank}(A) = n$

6.2 Analysis: Mon/Tue

6.2.1 Directional and Partial Derivatives

Definition 6.6 · Directional Derivative

Let $f : U \to \mathbb{R}^m$ where $U \subseteq \mathbb{R}^n$ is open, $a \in U$, $v \in \mathbb{R}^n$. The **directional derivative** of f at a in direction v is

$$(D_v f)(a) = \lim_{t \to 0} \frac{f(a + tv) - f(a)}{t}$$

if this limit exists.

Remark. This is the derivative of the single-variable function $g(t) = f(a + tv)$ at $t = 0$. When $v = e_j = (0, \ldots, 1, \ldots, 0)$, the directional derivative is called a **partial derivative**, denoted $(D_j f)(a)$ or $\frac{\partial f}{\partial x_j}(a)$.

Theorem 6.7 · Directional Derivative via Derivative Matrix

Suppose f is differentiable at $a \in U$. Then for any $v \in \mathbb{R}^n$, the directional derivative $(D_v f)(a)$ exists and

$$(D_v f)(a) = [Df(a)]v$$

Proof. Let $\varepsilon > 0$. We WTS $\left\| \frac{f(a+tv)-f(a)}{t} - Df(a) \cdot v \right\| < \varepsilon$ if $0 < |t| < \delta$.
If $v = 0$, this is obvious. Assume $v \neq 0$.

By definition of $Df(a)$, there exists $\delta' > 0$ such that

$$\frac{\|f(a+h) - f(a) - Df(a) \cdot h\|}{\|h\|} < \frac{\varepsilon}{\|v\|} \text{ whenever } \|h\| < \delta'$$

Let $\delta = \frac{\delta'}{\|v\|}$. If $0 < |t| < \delta$, then $h = tv$ satisfies $0 < \|h\| = |t|\|v\| < \delta'$.
Hence:

$$\frac{\|f(a+tv) - f(a) - Df(a) \cdot tv\|}{|t|\|v\|} < \frac{\varepsilon}{\|v\|}$$

which gives

$$\left\| \frac{f(a+tv) - f(a)}{t} - Df(a) \cdot v \right\| < \varepsilon$$

■

Theorem 6.8 · Corollary: Partial Derivatives

If f is differentiable at a:

1. $(D_j f)(a) = Df(a) \cdot e_j =$ *the j-th column of* $Df(a)$.
2. *For* $v = \sum_{j=1}^n v_j e_j$: $(D_v f)(a) = \sum_{j=1}^n v_j (D_j f)(a)$.

Theorem 6.9 · Continuous Partials Imply Differentiability

Given $U \subseteq \mathbb{R}^n$ open, $f : U \to \mathbb{R}^m$. Suppose there exists $a \in U$ and $\rho > 0$ such that $D_j f$ exists everywhere in $B_\rho(a)$ for all j, and $D_j f$ is continuous at a. Then f is differentiable at a.

Proof. Let $\varepsilon > 0$. We WTS

$$\frac{\left\| f(a+h) - f(a) - \sum_{j=1}^n (D_j f)(a) h_j \right\|}{\|h\|} < \varepsilon$$

whenever $0 < \|h\| < \delta$, where $h = \sum_{j=1}^n h_j e_j$.
Define $h^{(k)} = \sum_{j=1}^k h_j e_j$ for $k = 1, \ldots, n$.
Then:

$$f(a+h) - f(a) = \sum_{k=1}^n \left[f(a + h^{(k)}) - f(a + h^{(k-1)}) \right]$$

By MVT applied to each term:

$$f(a + h^{(k)}) - f(a + h^{(k-1)}) = h_k (D_k f)(a + h^{(k-1)} + \theta_k h_k e_k)$$

for some $\theta_k \in (0, 1)$.
Hence:

$$\frac{\left\| f(a+h) - f(a) - \sum_{j=1}^n (D_j f)(a) h_j \right\|}{\|h\|}$$

$$= \frac{\left\| \sum_{j=1}^n h_j \left[(D_j f)(a + h^{(j-1)} + \theta_j h_j e_j) - (D_j f)(a) \right] \right\|}{\|h\|}$$

$$\leq \sum_{j=1}^{n} \left\| (D_j f)(a + h^{(j-1)} + \theta_j h_j e_j) - (D_j f)(a) \right\|$$

Since $D_j f$ is continuous at a, there exists δ_j such that each term is $< \frac{\varepsilon}{n}$ when $\|h\| < \delta_j$. Take $\delta = \min\{\delta_1, \ldots, \delta_n\}$. ■

6.2.2 The Chain Rule

Theorem 6.10 · Chain Rule

Let $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ be open. Let $f : U \to V$ and $g : V \to \mathbb{R}^\ell$. Suppose f is differentiable at $a \in U$ and g is differentiable at $f(a) \in V$. Then $g \circ f : U \to \mathbb{R}^\ell$ is differentiable at a, and

$$\underbrace{D(g \circ f)(a)}_{\mathbb{R}^{\ell \times n}} = \underbrace{Dg(f(a))}_{\mathbb{R}^{\ell \times m}} \cdot \underbrace{Df(a)}_{\mathbb{R}^{m \times n}}$$

Proof. Let $\varepsilon > 0$. We WTS there exists $\delta > 0$ such that $B_\delta(a) \subseteq U$ and

$$\|(g \circ f)(a+h) - (g \circ f)(a) - Dg(f(a))Df(a)h\| < \varepsilon\|h\|$$

whenever $\|h\| < \delta$.
Let $\varepsilon_1, \varepsilon_2 > 0$ to be chosen. By differentiability of f at a, there exists $\delta_1 > 0$ such that

$$\|f(a+h) - f(a) - Df(a)h\| < \varepsilon_1\|h\| \text{ for } \|h\| < \delta_1$$

By differentiability of g at $f(a)$, there exists $\delta_2 > 0$ such that

$$\|g(y) - g(f(a)) - Dg(f(a))(y - f(a))\| < \varepsilon_2\|y - f(a)\| \text{ for } \|y - f(a)\| < \delta_2$$

Since f is continuous at a, there exists $\delta_3 > 0$ such that $\|f(a+h) - f(a)\| < \delta_2$ for $\|h\| < \delta_3$. For $\|h\| < \min\{\delta_1, \delta_3\}$:

$$\begin{aligned}
&\|(g \circ f)(a+h) - (g \circ f)(a) - Dg(f(a))Df(a)h\| \\
&\leq \|g(f(a+h)) - g(f(a)) - Dg(f(a))(f(a+h) - f(a))\| \\
&\quad + \|Dg(f(a))\|\|f(a+h) - f(a) - Df(a)h\| \\
&< \varepsilon_2\|f(a+h) - f(a)\| + \|Dg(f(a))\|\varepsilon_1\|h\| \\
&\leq (\varepsilon_1\varepsilon_2 + \varepsilon_2\|Df(a)\| + \varepsilon_1\|Dg(f(a))\|)\|h\|
\end{aligned}$$

Choose $\varepsilon_1, \varepsilon_2$ small enough that this is $< \varepsilon\|h\|$. ■

Theorem 6.11 · Chain Rule: Component Form

With f, g as above:

$$D_i(g \circ f)(a) = \sum_{k=1}^{m} (D_k g_i)(f(a)) \cdot (D_i f_k)(a)$$

Theorem 6.12 · One-dimensional Chain Rule

If $n = m = 1$*:* $(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$.

6.3 Linear Algebra: Thu/Fri

Content from Linear Algebra PDF needed

Week 7

Higher Derivatives and Taylor's Theorem

7.1 Analysis: Mon/Tue

7.1.1 Higher Derivatives

Given $f : U \to \mathbb{R}^m$ where $U \subseteq \mathbb{R}^n$ is open, we can take partial derivatives $D_i f$.

Definition 7.1 · Higher Partial Derivatives

Suppose $D_i f$ is defined everywhere on U. Then we can think of $D_i f : U \to \mathbb{R}^m$. Define

$$D_j D_i f(a) = D_j(D_i f)(a)$$

if it exists. Define higher derivatives inductively by

$$D_{i_1} D_{i_2} \cdots D_{i_k} f = D_{i_1}(D_{i_2} \cdots D_{i_k} f)$$

if $D_{i_2} \cdots D_{i_k} f : U \to \mathbb{R}^m$ is a well-defined function and this exists.

Definition 7.2 · C^k Functions

We say $f : U \to \mathbb{R}^m$ is C^k if f and all higher derivatives up to k-th order are well-defined and continuous.

Theorem 7.1 · Symmetry of Mixed Partials

Let $f : U \to \mathbb{R}^m$, $U \subseteq \mathbb{R}^n$ open. Suppose $a \in U$, $\rho > 0$ s.t. $B_\rho(a) \subseteq U$ and $D_i f$, $D_j f$, $D_i D_j f$, $D_j D_i f$ are defined in $B_\rho(a)$.
Assume, moreover, that $D_i D_j f$ and $D_j D_i f$ are continuous at a. Then

$$D_i D_j f(a) = D_j D_i f(a)$$

Sketch. Consider $\Delta = f(a + he_i + ke_j) - f(a + he_i) - f(a + ke_j) + f(a)$.
Define $\psi_k(x) = f(x + ke_j) - f(x)$. Then $\Delta = \psi_k(a + he_i) - \psi_k(a)$.

By MVT: $\Delta = D_i\psi_k(a+\theta he_i)\cdot h = [D_if(a+he_i+\theta' ke_j) - D_if(a+\theta' ke_j)]\cdot h$
$= D_jD_if(a+\theta'' he_i+\theta' ke_j)\cdot hk$ for some $\theta,\theta',\theta''\in(0,1)$.
Similarly, defining $\phi_h(x) = f(x+he_i)-f(x)$, we get $\Delta = D_iD_jf(a+\theta''' he_i+\theta'''' ke_j)\cdot hk$.
Taking $h,k\to 0$ and using continuity gives the result. ∎

Corollary 7.2 · Order Independence for C^k

Suppose $f: U\to\mathbb{R}^m$ is C^k, $k\geq 2$. Then

$$D_{i_1}\cdots D_{i_k}f(x) = D_{j_1}\cdots D_{j_k}f(x)$$

whenever $(j_1,\dots,j_k)$ is a permutation of $(i_1,\dots,i_k)$.

7.1.2 Taylor's Theorem

Definition 7.3 · Gradient and Hessian

1. For $f: U\to\mathbb{R}$ differentiable, the **gradient** is $\nabla f = (D_1f,\dots,D_nf)$.
2. A function $Q:\mathbb{R}^n\to\mathbb{R}$ is called a **quadratic form** if $Q(x) = \sum_{i,j} h_{ij}x_ix_j$ for some symmetric matrix (h_{ij}).
3. The quadratic form $\sum_{i,j} D_iD_jf(a)\cdot x_ix_j$ is called the **Hessian** of f at a.
4. Q is **positive definite** if $Q(x)>0$ for all $x\neq 0$.

Theorem 7.3 · Second-Order Taylor Expansion

Let $U\subseteq\mathbb{R}^n$ open, $f: U\to\mathbb{R}$ be C^2. Then for any $a\in U$:

$$f(x) = f(a) + \nabla f(a)\cdot(x-a) + \frac{1}{2}\sum_{i,j} D_iD_jf(a)(x-a)_i(x-a)_j + o(\|x-a\|^2)$$

where $\lim_{x\to a}\frac{o(\|x-a\|^2)}{\|x-a\|^2} = 0$.

7.1.3 Second Derivative Test

Lemma 7.4 · Critical Points

Let U open, $f: U\to\mathbb{R}$ differentiable. Suppose f has a local min/max at a. Then $\nabla f(a) = 0$.

Theorem 7.5 · Second Derivative Test

Let U open, $f: U\to\mathbb{R}$ be C^2. Suppose $\nabla f(a) = 0$ and the Hessian of f at a is positive definite. Then f has a strict local minimum at a.

Sketch. **Lemma:** If Q is a positive definite quadratic form, then $\exists M > 0$ s.t. $Q(x) \geq M\|x\|^2$ for all $x \in \mathbb{R}^n$.
Assuming the lemma, then $\exists \rho$ s.t. for $x \in B_\rho(a)$:

$$f(x) \approx f(a) + \frac{1}{2}\text{Hess}_f(x-a) + o(\|x-a\|^2) \geq f(a) + \frac{M}{2}\|x-a\|^2 - o(\|x-a\|^2)$$

For small enough $\|x-a\|$, this is $> f(a)$. ∎

7.2 Linear Algebra: Thu/Fri (Orthonormal Bases)

7.2.1 Orthonormal Bases

Definition 7.4 · Orthonormal Set

Vectors $\{u_1, \dots, u_k\} \subseteq V$ are **orthonormal** if

$$\langle u_i, u_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

An **orthonormal basis** is a basis that is orthonormal.

Lemma 7.6 · Orthonormal Implies Independent

If $\{u_1, \dots, u_k\}$ is orthonormal, then it is linearly independent.

Proof. Suppose $\sum_{j=1}^k \alpha_j u_j = 0$. Take dot product with u_i:

$$0 = \sum_{j=1}^k \alpha_j \langle u_i, u_j \rangle = \alpha_i$$

Hence $\alpha_i = 0$ for all i. ∎

7.2.2 Orthogonal Projection

Definition 7.5 · Orthogonal Projection

For $W \subseteq \mathbb{F}^n$ a subspace and $x \in \mathbb{F}^n$, there exists a unique decomposition $x = w + w^\perp$ where $w \in W$ and $w^\perp \in W^\perp$.
Define $P_W : \mathbb{F}^n \to W$ by $P_W(x) = w$. This is the **orthogonal projection** onto W.

Theorem 7.7 · Projection Formula

Suppose $\{u_1, \dots, u_k\}$ is an orthonormal basis of W. Then

$$P_W(x) = \sum_{j=1}^k \langle x, u_j \rangle u_j$$

Proof. By definition, it suffices to show $v = x - \sum_j \langle x, u_j \rangle u_j \in W^\perp$.
Let $\langle v, u_i \rangle$ for any i. Since $\{u_1, \ldots, u_k\}$ is a basis, we need $\langle v, u_i \rangle = 0$ for all i.
$\langle v, u_i \rangle = \langle x, u_i \rangle - \sum_j \langle x, u_j \rangle \langle u_j, u_i \rangle = \langle x, u_i \rangle - \langle x, u_i \rangle = 0$. ∎

7.2.3 Gram-Schmidt Process

Theorem 7.8 · Gram-Schmidt

Suppose $\{v_1, \ldots, v_k\} \subseteq V$ is linearly independent. Then there exists an orthonormal set $\{u_1, \ldots, u_k\}$ s.t.

$$\mathit{span}(u_1, \ldots, u_j) = \mathit{span}(v_1, \ldots, v_j)$$

for any $j = 1, \ldots, k$.

Proof. Define $u_1 = \frac{v_1}{\|v_1\|}$ (note $v_1 \neq 0$ since linearly independent). Moreover, $\{u_1\}$ is orthonormal and $\text{span}(u_1) = \text{span}(v_1)$.
Inductively, suppose we have $\{u_1, \ldots, u_j\}$ orthonormal with the right span.
Define $\tilde{u}_{j+1} = v_{j+1} - P_{\text{span}(u_1, \ldots, u_j)}(v_{j+1}) = v_{j+1} - \sum_{i=1}^{j} \langle v_{j+1}, u_i \rangle u_i$.
Then $\tilde{u}_{j+1} \neq 0$ (otherwise $v_{j+1} \in \text{span}(v_1, \ldots, v_j)$, contradicting independence).
Set $u_{j+1} = \frac{\tilde{u}_{j+1}}{\|\tilde{u}_{j+1}\|}$. ∎

7.2.4 Eigenvalues and the Spectral Theorem

Now we come to one of the most important results in linear algebra: the Spectral Theorem for symmetric matrices.

Definition 7.6 · Characteristic Polynomial

Let A be an $n \times n$ matrix. The **characteristic polynomial** of A is

$$p_A(\lambda) = \det(A - \lambda I)$$

This is a degree n polynomial in λ, with leading coefficient $(-1)^n$.

Definition 7.7 · Eigenvalue and Eigenvector

Let A be an $n \times n$ matrix.

1. The **eigenvalues** of A are the roots of the characteristic polynomial, i.e., values λ such that $\det(A - \lambda I) = 0$.
2. An **eigenvector** corresponding to eigenvalue λ is a nonzero vector $v \in \mathbb{R}^n$ such that $Av = \lambda v$.

Remark. By the Fundamental Theorem of Algebra, every $n \times n$ matrix has exactly n eigenvalues (counting multiplicity), though some may be complex. We're mainly interested in the case where eigenvalues are real.

Lemma 7.9 · Properties of Eigenvalues

Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$. Then:

1. $\det A = \lambda_1 \cdots \lambda_n$ *(product of eigenvalues)*
2. *For $\lambda \in \mathbb{R}$: λ is an eigenvalue $\iff \exists v \neq 0$ with $Av = \lambda v$*

Proof. For (1): Set $\lambda = 0$ in $\det(A - \lambda I) = (-1)^n(\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$.
For (2): $\det(A - \lambda I) = 0 \iff N(A - \lambda I) \neq \{0\} \iff \exists v \neq 0$ with $(A - \lambda I)v = 0 \iff Av = \lambda v$. ∎

Lemma 7.10 · Eigenvalues of Symmetric Matrices are Real

If A is a real symmetric matrix, then all eigenvalues of A are real.

Proof. Suppose $Av = \lambda v$ for some $v \neq 0$ (possibly complex). Taking the conjugate transpose:

$$\bar{v}^T A = \bar{\lambda} \bar{v}^T$$

since $A = A^T$ and A is real. Multiplying the original equation by $\bar{v}^T$:

$$\bar{v}^T A v = \lambda \bar{v}^T v = \lambda \|v\|^2$$

But also $\bar{v}^T A v = \bar{\lambda} \bar{v}^T v = \bar{\lambda} \|v\|^2$.
Since $\|v\|^2 \neq 0$, we get $\lambda = \bar{\lambda}$, so λ is real. ∎

Theorem 7.11 · Spectral Theorem

Let A be a real symmetric $n \times n$ matrix. Then there exists an orthonormal basis $\{v_1, \dots, v_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A.

Proof. We proceed by induction on n.
Base case: $n = 1$ is trivial — A is just a scalar and $\mathbb{R}^1$ has the standard basis.
Inductive step: Assume the theorem holds for $(n-1) \times (n-1)$ symmetric matrices.
Step 1: Find a first eigenvector.
Consider the quadratic form $Q(x) = x^T A x = \sum_{i,j} a_{ij} x_i x_j$.
Restricted to the unit sphere S^{n-1}, Q is continuous on a compact set, so it attains a minimum at some point $v_1 \in S^{n-1}$.
Since $Q(x) = \|x\|^2 Q(\hat{x})$ where $\hat{x} = x/\|x\|$, the function $\|x\|^{-2} Q(x)$ also attains its minimum at v_1.
Taking the gradient and setting it to zero (this is a constrained optimization on the sphere, using Lagrange multipliers), we find that

$$Av_1 = \lambda_1 v_1$$

where $\lambda_1 = Q(v_1)$ is the minimum value. So v_1 is an eigenvector!
Step 2: Restrict to the orthogonal complement.

Let $V = \text{span}\{v_1\}$ and $V^\perp$ be its orthogonal complement. Since $\dim V = 1$, we have $\dim V^\perp = n - 1$.
Claim: A maps $V^\perp$ to itself, i.e., if $w \perp v_1$, then $Aw \perp v_1$.
Proof of claim: For $w \in V^\perp$:

$$\langle Aw, v_1 \rangle = \langle w, A^T v_1 \rangle = \langle w, Av_1 \rangle = \langle w, \lambda_1 v_1 \rangle = \lambda_1 \langle w, v_1 \rangle = 0$$

using that $A = A^T$ (symmetric).
Step 3: Apply induction.
Choose an orthonormal basis $\{w_1, \ldots, w_{n-1}\}$ for $V^\perp$. The restriction $A|_{V^\perp} : V^\perp \to V^\perp$ is represented by some $(n-1) \times (n-1)$ matrix A_0 relative to this basis.
One can check that A_0 is also symmetric (same argument using $A = A^T$).
By the inductive hypothesis, there exists an orthonormal basis $\{u_2, \ldots, u_n\}$ of $\mathbb{R}^{n-1}$ consisting of eigenvectors of A_0:

$$A_0 u_j = \lambda_j u_j \text{ for } j = 2, \ldots, n$$

Define $v_j = \sum_{i=1}^{n-1} (u_j)_i w_i \in V^\perp$ for $j = 2, \ldots, n$.
Then $\{v_1, v_2, \ldots, v_n\}$ is an orthonormal basis for $\mathbb{R}^n$ (since $v_1 \perp V^\perp$ and $\{v_2, \ldots, v_n\}$ is orthonormal in $V^\perp$), and each v_j is an eigenvector of A. ■

Corollary 7.12 · Diagonalization of Symmetric Matrices

If A is a real symmetric matrix, then there exists an orthogonal matrix Q such that $Q^T AQ$ is diagonal. The diagonal entries are the eigenvalues of A.

Proof. Let $\{v_1, \ldots, v_n\}$ be the orthonormal eigenvector basis from the Spectral Theorem. Let Q be the matrix with columns $v_1, \ldots, v_n$. Then Q is orthogonal and

$$Q^T AQ = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

■

Week 8

Integration and the Fundamental Theorem

8.1 Analysis: Mon/Tue

8.1.1 The Riemann Integral

Definition 8.1 · Riemann Integral

For $f : [a,b] \to \mathbb{R}$:

1. A **partition** P is a choice $a = t_0 < t_1 < \cdots < t_n = b$.
2. Define
$$L(f,P) = \sum_{i=1}^{n} \inf_{t\in[t_{i-1},t_i]} f(t)\cdot(t_i - t_{i-1})$$
$$U(f,P) = \sum_{i=1}^{n} \sup_{t\in[t_{i-1},t_i]} f(t)\cdot(t_i - t_{i-1})$$
3. Define $\underline{\int} f = \sup_P L(f,P)$ and $\overline{\int} f = \inf_P U(f,P)$.
4. f is **integrable** if $\underline{\int} f = \overline{\int} f$, and we write this value as $\int_a^b f(t)\,dt$.

Theorem 8.1 · Properties of the Integral

Let $f,g : [a,b] \to \mathbb{R}$ be integrable and $c_1, c_2 \in \mathbb{R}$. Then:

1. *$c_1 f + c_2 g$ is integrable and $\int(c_1 f + c_2 g) = c_1 \int f + c_2 \int g$*
2. *(**Positivity**) If $f \geq 0$, then $\int f \geq 0$*
3. *(**Boundedness**) $|\int f| \leq \sup|f|\cdot(b-a)$*
4. *(**Additivity**) If $a < c < b$, then $\int_a^b f = \int_a^c f + \int_c^b f$*

8.1.2 Fundamental Theorem of Calculus

Theorem 8.2 · FTC Version 1

Suppose $f : [a, b] \to \mathbb{R}$ is continuous. Define $F : [a, b] \to \mathbb{R}$ by

$$F(t) = \int_a^t f(s)\, ds$$

Then F is C^1 and $F'(t) = f(t)$ for all $t \in (a, b)$.

Proof. Suffices to show $F'(t) = f(t)$ (then f cts implies F' cts).
Let $\varepsilon > 0$. Consider $|h| > 0$.
$\left|\frac{F(t+h)-F(t)}{h} - f(t)\right| = \left|\frac{1}{h}\int_t^{t+h} f(s)\, ds - f(t)\right| = \left|\frac{1}{h}\int_t^{t+h}(f(s) - f(t))\, ds\right|$
By continuity of f, $\exists\delta > 0$ s.t. $|f(s) - f(t)| < \varepsilon$ whenever $|s - t| < \delta$.
If $|h| < \delta$:

$$\left|\frac{1}{h}\int_t^{t+h}(f(s) - f(t))\, ds\right| \leq \frac{1}{|h|} \cdot \varepsilon \cdot |h| = \varepsilon$$

■

Theorem 8.3 · FTC Version 2

Suppose $f : [a, b] \to \mathbb{R}$ is C^1. Then

$$\int_a^b f'(t)\, dt = f(b) - f(a)$$

Proof. Define $F(t) = \int_a^t f'(s)\, ds$. By FTC v1, F is C^1 and $F'(t) = f'(t)$.
Let $g(t) = F(t) - f(t)$. Then g is C^1 and $g'(t) = F'(t) - f'(t) = 0$ for all t.
Lemma: If f is continuous on $[a, b]$, differentiable on (a, b), and $f'(t) = 0$ for all t, then f is constant.
Hence g is constant, so $F(a) - f(a) = g(a) = g(b) = F(b) - f(b)$.
Rearranging: $f(b) - f(a) = F(b) - F(a) = F(b) = \int_a^b f'(s)\, ds$. ■

8.1.3 Integration by Parts

Theorem 8.4 · Integration by Parts

Let $f, g : [a, b] \to \mathbb{R}$ be C^1. Then

$$\int_a^b f'(t)g(t)\, dt = f(b)g(b) - f(a)g(a) - \int_a^b f(t)g'(t)\, dt$$

Proof. Apply FTC v2 to $fg : [a, b] \to \mathbb{R}$:

$$f(b)g(b) - f(a)g(a) = \int_a^b (fg)'(t)\, dt = \int_a^b f'g\, dt + \int_a^b fg'\, dt$$

Rearranging gives the result. ■

8.2 Linear Algebra: Thu/Fri (Computing Inverses)

8.2.1 Finding Inverses via Row Reduction

Theorem 8.5 · Inverse via Augmented Matrix

Augment $(A \mid I)$ and row reduce. If rref$(A) = I$, then rref$(A \mid I) = (I \mid A^{-1})$.

Proof. Recall that row operations correspond to multiplying by matrices on the left. Hence if $R_1, \ldots, R_k$ are the row operations:

$$\text{rref}(A \mid I) = R_k \cdots R_1(A \mid I) = (R_k \cdots R_1 A \mid R_k \cdots R_1 I)$$

Here $B = R_k \cdots R_1 = A^{-1}$. ■

Remark. The row operations correspond to matrices:

- Switching rows i and j: permutation matrix
- Multiply row i by $\lambda \neq 0$: diagonal matrix with λ in position (i, i)
- Add $\lambda \times$ row j to row i: elementary matrix with λ in position (i, j)

All of these are invertible!

Week 9

Curves and Submanifolds

9.1 Analysis: Mon/Tue

9.1.1 Curves

Definition 9.1 · Curve

A **curve** in $\mathbb{R}^m$ is a map $\gamma : [a, b] \to \mathbb{R}^m$.
We say γ is a C^k curve for $k \in \mathbb{N} \cup \{\infty\}$ if each component function is C^k.

Definition 9.2 · Velocity and Unit Tangent

1. If $\gamma : [a, b] \to \mathbb{R}^m$ is C^1, define the **velocity vector** to be $\gamma'(t) \in \mathbb{R}^m$.
2. If $\gamma'(t) \neq 0$, define the **unit tangent vector** to be $T(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$.

9.1.2 Arc Length

Definition 9.3 · Length of a Curve

For a partition $P : a = t_0 < t_1 < \cdots < t_n = b$, define

$$L(\gamma, P) = \sum_{i=1}^{n} \|\gamma(t_i) - \gamma(t_{i-1})\|$$

We say γ has **finite length** if $\{L(\gamma, P) : P \text{ a partition}\}$ is bounded above.
If γ has finite length, define $L(\gamma) = \sup_P L(\gamma, P)$.

Theorem 9.1 · Arc Length Formula

Suppose $\gamma : [a,b] \to \mathbb{R}^m$ is C^1. Then γ has finite length and

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt$$

Sketch. **Step 1:** γ is of finite length.
For any partition P:

$$L(\gamma, P) = \sum \|\gamma(t_i) - \gamma(t_{i-1})\| = \sum \left\| \int_{t_{i-1}}^{t_i} \gamma'(s) \, ds \right\| \leq \sum \int_{t_{i-1}}^{t_i} \|\gamma'(s)\| \, ds = \int_a^b \|\gamma'(s)\| \, ds$$

Hence $\{L(\gamma, P)\}$ is bounded above by $\int_a^b \|\gamma'\| \, dt$.
Step 2: Define $S(t) = L(\gamma|_{[a,t]})$.
Claim: $S'(t)$ exists and $S'(t) = \|\gamma'(t)\|$.
Using FTC: $S(b) = S(a) + \int_a^b S'(t) \, dt = 0 + \int_a^b \|\gamma'(t)\| \, dt$. ∎

Lemma 9.2 · Additivity of Length

$\gamma : [a,b] \to \mathbb{R}^m$ has finite length. Suppose $c \in (a,b)$. Then $\gamma|_{[a,c]}$ and $\gamma|_{[c,b]}$ have finite lengths and

$$L(\gamma) = L(\gamma|_{[a,c]}) + L(\gamma|_{[c,b]})$$

9.1.3 Submanifolds

Definition 9.4 · Graph Map

Let $U \subseteq \mathbb{R}^k$ be open, $g : U \to \mathbb{R}^{n-k}$ be C^1.
Define the **graph map** $\Gamma : U \to \mathbb{R}^n$ by

$$\Gamma(x) = (x, g(x)) = (x_1, \ldots, x_k, g_1(x_1, \ldots, x_k), \ldots, g_{n-k}(x_1, \ldots, x_k))$$

Definition 9.5 · k-Dimensional Submanifold

We say $M \subseteq \mathbb{R}^n$ is a **k-dimensional C^1 submanifold** if for all $a \in M$, $\exists \rho > 0$ s.t.

$$M \cap B_\rho(a) = P_{j_1,\ldots,j_k}^{-1}(\Gamma(U))$$

where:

- Γ is the graph map for some C^1 function $g : U \to \mathbb{R}^{n-k}$, $U \subseteq \mathbb{R}^k$ open
- $P_{j_1,\ldots,j_k}$ is a permutation of coordinates

I.e., $M \cap B_\rho(a)$ can be represented as a graph after permuting coordinates.

Theorem 9.3 · Level Set Theorem

Suppose $\phi_1, \dots, \phi_{n-k} : \mathbb{R}^n \to \mathbb{R}$ are C^1.
Let $M = \{x \in \mathbb{R}^n : \phi_1(x) = \cdots = \phi_{n-k}(x) = 0\}$.
If $\nabla\phi_1(x), \dots, \nabla\phi_{n-k}(x)$ are linearly independent for all $x \in M$, then (if $M \neq \emptyset$) M is a k-dimensional C^1 submanifold.

Example: $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$.
Here $\phi(x) = \|x\|^2 - 1$. We have $\nabla\phi(x) = 2x$, which is linearly independent when $\phi(x) = 0$ (i.e., $x \neq 0$).
So S^{n-1} is an $(n-1)$-dimensional submanifold.

9.1.4 Tangent Spaces

Definition 9.6 · Tangent Space

Let $M \subseteq \mathbb{R}^n$ be a k-dimensional submanifold, $a \in M$.
Define the **tangent space** of M at a by

$$T_a M = \{\gamma'(t_0) : \gamma : (-\varepsilon, \varepsilon) \to M \text{ is a } C^1 \text{ curve s.t. } \gamma(t_0) = a\}$$

Theorem 9.4 · Tangent Space is a Subspace

$T_a M$ is a k-dimensional linear subspace of $\mathbb{R}^n$.

Sketch. Suffices to consider the case of a graph. If $M \cap B_\rho(a) = \Gamma(U)$ where $\Gamma(x) = (x, g(x))$, then:
Claim: $T_a M = \operatorname{span}(D_1\Gamma(a), \dots, D_k\Gamma(a))$.
Here $D_i\Gamma(a) = (e_i, D_i g(a))$.
These are linearly independent (first k components are $e_1, \dots, e_k$), so $\dim T_a M = k$. ■

9.2 Linear Algebra: Thu/Fri (Abstract Vector Spaces)

Remark. Many results from $\mathbb{R}^n$ extend to abstract vector spaces over any field $\mathbb{F}$. The key tools:

- Bases and dimension
- Linear maps and their matrices
- Kernel, range, rank-nullity theorem
- Change of basis

For inner product spaces specifically (over $\mathbb{R}$ or $\mathbb{C}$):

- Orthogonality and orthonormal bases

- Gram-Schmidt process
- Orthogonal projections

Week 10

Power Series and Integration in Higher Dimensions

10.1 Analysis: Mon/Tue

10.1.1 Power Series

Consider a power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ where the coefficients $a_n \in \mathbb{R}$.

Lemma 10.1 · Convergence Propagates

Suppose $\exists x_0 \neq 0$ s.t. $\sum a_n x_0^n$ converges. Then:

1. *For all x with $|x| < |x_0|$, the power series converges absolutely.*
2. *For all $R' < |x_0|$ and any polynomial $p(n)$, $\sum p(n) a_n x^n$ converges absolutely and uniformly on $[-R', R']$.*

Sketch. Let $r = |x|/|x_0| < 1$.
Since $\sum a_n x_0^n$ converges, $\lim_{n\to\infty} a_n x_0^n = 0$. So $\exists B > 0$ s.t. $|a_n x_0^n| \leq B$ for all n.
Hence $|a_n x^n| = |a_n x_0^n| \cdot r^n \leq Br^n$.
Since $\sum Br^n$ converges (geometric series with $r < 1$), we get absolute convergence.
For uniform convergence, note that $|p(n)| \cdot |a_n| \cdot |x|^n \leq |p(n)|Br^n$ on $[-R', R']$, and $\sum |p(n)|r^n$ converges. ∎

Theorem 10.2 · Radius of Convergence

For any power series $\sum a_n x^n$, exactly one of the following holds:

1. *$\forall x \in \mathbb{R}$, $f(x)$ is convergent (radius $R = \infty$)*
2. *$f(x)$ is divergent $\forall x \neq 0$ (radius $R = 0$)*
3. *$\exists R > 0$ s.t. $f(x)$ is convergent if $|x| < R$ and divergent if $|x| > R$*

In case (3), nothing is claimed for $x = \pm R$.

Definition 10.1 · Radius of Convergence

The **radius of convergence** is ∞ in case (1), 0 in case (2), and R in case (3).

10.1.2 Uniform Convergence

Definition 10.2 · Uniform Convergence

Let $f_j : S \to \mathbb{R}$. We say f_j **converges to** f **uniformly** if

$$\lim_{j\to\infty} \sup_{x\in S} |f_j(x) - f(x)| = 0$$

I.e., $\forall \varepsilon > 0$, $\exists N$ s.t. $|f_j(x) - f(x)| < \varepsilon$ for all $j \geq N$ and all $x \in S$.

Theorem 10.3 · Uniform Limit of Continuous Functions

Suppose $f_j : S \to \mathbb{R}$ are continuous and $f_j \to f$ uniformly. Then f is continuous.

Theorem 10.4 · Weierstrass M-Test

Suppose $\sum g_j : S \to \mathbb{R}$ satisfies $|g_j(x)| \leq M_j$ for some M_j with $\sum M_j$ convergent.
Then $\sum g_j$ converges uniformly to some g.
In particular, if each g_j is continuous, then g is also continuous.

Theorem 10.5 · Continuity of Power Series

Let $f(x) = \sum a_n x^n$ have radius of convergence $R > 0$. Then f is continuous on $(-R, R)$.

Theorem 10.6 · Differentiability of Power Series

Let $f(x) = \sum a_n x^n$ have radius of convergence $R > 0$. Then:

1. *f is C^k on $(-R, R)$ for all k*
2. *$f^{(k)}(x) = \sum_{n\geq k} n(n-1)\cdots(n-k+1)a_n x^{n-k}$*

This means we can differentiate term by term!

10.1.3 Taylor Series

Theorem 10.7 · Taylor's Theorem (General)

Let $f : [x - r, x + r] \to \mathbb{R}$ be C^{k+1} for some $L \in \mathbb{R}$, $r > 0$. Then

$$f(x) = \sum_{j=0}^{k} \frac{f^{(j)}(L)}{j!}(x - L)^j + R_k(x)$$

where the remainder satisfies $R_k(x) = \frac{f^{(k+1)}(\xi)}{(k+1)!}(x - L)^{k+1}$ for some ξ between L and x.

Corollary 10.8 · Power Series Representation

If $f : (L - r, L + r) \to \mathbb{R}$ is C^∞ and $|f^{(k)}(x)| \to 0$ sufficiently fast, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(L)}{n!}(x - L)^n$$

In particular, $a_n = \frac{f^{(n)}(L)}{n!}$.

10.1.4 Integration in $\mathbb{R}^n$

Definition 10.3 · Rectangle and Partition

1. A **rectangle** $R = R[a_1, b_1; \ldots; a_n, b_n] = \{(x_1, \ldots, x_n) : a_i \leq x_i \leq b_i\}$
2. $\text{Vol}(R) = (b_1 - a_1) \cdots (b_n - a_n)$
3. A **partition** P of R is a choice of partitions P_i of $[a_i, b_i]$ for each i

Definition 10.4 · Riemann Integral in $\mathbb{R}^n$

For $f : R \to \mathbb{R}$ bounded:

$$L(f, P) = \sum_{R' \text{ subrectangle}} \inf_{x \in R'} f(x) \cdot \text{Vol}(R')$$

$$U(f, P) = \sum_{R' \text{ subrectangle}} \sup_{x \in R'} f(x) \cdot \text{Vol}(R')$$

Define $\underline{\int} f = \sup_P L(f, P)$, $\overline{\int} f = \inf_P U(f, P)$.
f is **integrable** if $\underline{\int} f = \overline{\int} f$, written $\int_R f$.

Theorem 10.9 · Fubini's Theorem

Given $f : R \to \mathbb{R}$ integrable, $R \subseteq \mathbb{R}^n$ a rectangle, $R = R[a_1, b_1; \dots; a_n, b_n]$.
For $x \in \mathbb{R}^n$, write $x = (y, z)$ where $y \in \mathbb{R}^k$ and $z \in \mathbb{R}^{n-k}$.
Then

$$\int_R f \, dx_1 \cdots dx_n = \int_{R'} \left(\int_{R''} f(y,z) \, dz \right) dy$$

where R' and R'' are the projections of R onto the y and z coordinates.
I.e., we can compute higher-dimensional integrals as iterated 1-D integrals in any order!

10.2 Linear Algebra: Thu/Fri (Review and Connections)

Remark. The main theorems connecting linear algebra and analysis in this course:

1. **Derivative as a linear map:** $(Df)(a) : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, represented by the Jacobian matrix.
2. **Chain rule as matrix multiplication:** $D(g \circ f)(a) = Dg(f(a)) \cdot Df(a)$.
3. **Tangent spaces as subspaces:** $T_a M$ is a linear subspace of $\mathbb{R}^n$.
4. **Hessian as a quadratic form:** The second derivative test uses positive definiteness of the Hessian matrix.
5. **Inverse Function Theorem:** f is locally invertible near a iff $Df(a)$ is invertible (i.e., $\det Df(a) \neq 0$).

10.3 Advanced Analysis: Contraction Mapping and Function Theorems

10.3.1 Contraction Mapping Principle

Definition 10.5 · Contraction

Let $E \subseteq \mathbb{R}^n$. A map $f : E \to \mathbb{R}^n$ is a **contraction** if there exists $\theta \in (0,1)$ such that

$$\|f(x) - f(y)\| \leq \theta \|x - y\| \quad \forall x, y \in E$$

Theorem 10.10 · Contraction Mapping Theorem

Let $E \subseteq \mathbb{R}^n$ be closed and let $f : E \to E$ be a contraction (so $f(E) \subseteq E$). Then f has a unique fixed point, i.e., there exists a unique $z \in E$ such that $f(z) = z$.

Proof. The proof is constructive — we'll actually find the fixed point!
Pick any $x_0 \in E$ and define iteratively:

$$x_k = f(x_{k-1}) \text{ for } k \geq 1$$

Step 1: The sequence is Cauchy.
By the contraction property:

$$\|x_{k+1} - x_k\| = \|f(x_k) - f(x_{k-1})\| \leq \theta \|x_k - x_{k-1}\|$$

By induction: $\|x_{k+1} - x_k\| \leq \theta^k \|x_1 - x_0\|$.
For $\ell > k$:

$$\begin{aligned} \|x_\ell - x_k\| &\leq \sum_{j=k}^{\ell-1} \|x_{j+1} - x_j\| \\ &\leq \sum_{j=k}^{\ell-1} \theta^j \|x_1 - x_0\| \\ &\leq \frac{\theta^k}{1-\theta} \|x_1 - x_0\| \end{aligned}$$

Since $\theta^k \to 0$, the sequence is Cauchy.
Step 2: The limit is a fixed point.
Since E is closed and $\{x_k\}$ is Cauchy in E, it converges to some $z \in E$.
Since f is continuous (contractions are Lipschitz, hence continuous):

$$f(z) = f(\lim_{k\to\infty} x_k) = \lim_{k\to\infty} f(x_k) = \lim_{k\to\infty} x_{k+1} = z$$

Step 3: Uniqueness.
If $f(z) = z$ and $f(w) = w$, then $\|z - w\| = \|f(z) - f(w)\| \leq \theta \|z - w\|$.
Since $\theta < 1$, this forces $\|z - w\| = 0$, so $z = w$. ∎

Remark. The contraction mapping theorem is incredibly useful! It gives both existence and uniqueness, plus a practical algorithm (iteration) to find the fixed point.

10.3.2 Inverse Function Theorem

The Inverse Function Theorem tells us when a C^1 map is locally invertible.

Theorem 10.11 · Inverse Function Theorem

Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}^n$ be C^1, and $x_0 \in U$ with $Df(x_0)$ invertible. Then there exist open sets V, W with $x_0 \in V$ and $f(x_0) \in W$ such that:

1. *$f|_V : V \to W$ is a bijection*
2. *The inverse $g : W \to V$ is also C^1*
3. *$Dg(y) = (Df(g(y)))^{-1}$ for all $y \in W$*

Sketch. Step 1: Reduce to the case $Df(x_0) = I$.
If $Df(x_0) = A$ is invertible, consider $\tilde{f}(x) = A^{-1} f(x)$. Then $D\tilde{f}(x_0) = A^{-1}A = I$.

If the theorem holds for $\tilde{f}$, it holds for f.
Step 2: Setup.
Since Df is continuous and $Df(x_0) = I$, there exists $\rho > 0$ such that $\|Df(x) - I\| < \frac{1}{2}$ for $x \in B_\rho(x_0)$.
Step 3: Injectivity.
For $x, y \in B_\rho(x_0)$:

$$\|f(x) - f(y) - (x - y)\| = \left\| \int_0^1 (Df(x + t(y - x)) - I)(y - x)\, dt \right\| \leq \frac{1}{2}\|x - y\|$$

Hence $\|f(x) - f(y)\| \geq \frac{1}{2}\|x - y\|$, so f is injective on $B_\rho(x_0)$.
Step 4: Surjectivity (using contraction mapping).
For any v near $f(x_0)$, we want to find x with $f(x) = v$.
Define $F_v(x) = x - f(x) + v$. We show F_v is a contraction on a closed ball centered at x_0:
$\|F_v(x) - F_v(y)\| = \|(x - f(x)) - (y - f(y))\| \leq \frac{1}{2}\|x - y\|$.
By the Contraction Mapping Theorem, F_v has a fixed point x, meaning $x = x - f(x) + v$, i.e., $f(x) = v$.
Step 5: Differentiability of inverse.
The formula $Dg(y) = (Df(g(y)))^{-1}$ follows from differentiating $f(g(y)) = y$ using the chain rule. ■

10.3.3 Implicit Function Theorem

The Implicit Function Theorem is closely related to the IFT and tells us when we can solve equations locally.

Theorem 10.12 · Implicit Function Theorem

Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open and $F : U \to \mathbb{R}^m$ be C^1. Suppose $(a, b) \in U$ satisfies $F(a, b) = 0$ and the partial derivative $D_y F(a, b) : \mathbb{R}^m \to \mathbb{R}^m$ is invertible.
Then there exist open neighborhoods V of a in $\mathbb{R}^n$ and W of b in $\mathbb{R}^m$, and a unique C^1 function $g : V \to W$ such that:

1. $g(a) = b$
2. $F(x, g(x)) = 0$ *for all* $x \in V$
3. $Dg(x) = -(D_y F(x, g(x)))^{-1} D_x F(x, g(x))$

Sketch. Define $\Phi : U \to \mathbb{R}^n \times \mathbb{R}^m$ by $\Phi(x, y) = (x, F(x, y))$.
Then $D\Phi(a, b) = \begin{pmatrix} I & 0 \\ D_x F & D_y F \end{pmatrix}$, which is invertible since $D_y F$ is invertible.
By the Inverse Function Theorem, Φ has a local inverse $\Psi = (\Psi_1, \Psi_2)$.
For $(x, 0)$ near $(a, 0)$: $\Psi(x, 0) = (x, g(x))$ for some function g.
Then $\Phi(x, g(x)) = (x, 0)$ implies $F(x, g(x)) = 0$.
The formula for Dg follows from differentiating $F(x, g(x)) = 0$. ■

Remark. The IFT and Implicit FT are fundamental tools in differential geometry — they let us show that submanifolds defined by equations are "nice" smooth objects.

Solutions to Selected Exercises

Part I: Math 61

Week 1: The Real Numbers

1.1. (i) Assume for contradiction that $-a > 0$. Then $a + (-a) > 0$ (O2), so $0 > 0$ (F5), contradiction. Also $-a \neq 0$ since $a \neq 0$. Hence $-a < 0$.
(ii) Assume $\frac{1}{a} < 0$. Then $a \cdot (-\frac{1}{a}) > 0$ (O2). But $a \cdot (-\frac{1}{a}) = -1$, so $(-1)(-1) = 1 > 0$ implies $1 + (-1) > 0$, contradicting F5. Also $\frac{1}{a} \neq 0$ since $a \cdot \frac{1}{a} = 1 \neq 0$. Hence $\frac{1}{a} > 0$.
(iii) Assume $\frac{1}{b} < \frac{1}{a}$. Multiplying by $ab > 0$ (using O2 and part (ii)): $a < b$, contradiction. Hence $\frac{1}{a} < \frac{1}{b}$. □

1.3. Let $S = \{x : x^2 < a, x > 0\}$. First, S is nonempty (take $x = \min(1, a/2)$) and bounded above (by $\max(1, a)$). Let $\alpha = \sup S$.
Claim: $\alpha^2 = a$.
If $\alpha^2 > a$: Choose n large enough that $\frac{1}{n} < \frac{\alpha^2 - a}{2\alpha}$. Then $(\alpha - \frac{1}{n})^2 > a$, so $\alpha - \frac{1}{n}$ is an upper bound for S, contradicting $\alpha = \sup S$.
If $\alpha^2 < a$: Choose n large enough that $(\alpha + \frac{1}{n})^2 < a$. Then $\alpha + \frac{1}{n} \in S$, contradicting $\alpha = \sup S$.
Hence $\alpha^2 = a$, and $\alpha = \sqrt{a}$. □

1.5. Let $\varepsilon > 0$. By Archimedean property, $\exists N \in \mathbb{N}$ with $N > 1/\varepsilon$, i.e., $\frac{1}{N} < \varepsilon$. For $n \geq N$: $\frac{1}{n} \leq \frac{1}{N} < \varepsilon$, so $|\frac{1}{n} - 0| < \varepsilon$. Hence $\lim \frac{1}{n} = 0$. □

1.7. No! Note that $(v_1 - v_2) + (v_2 - v_3) + (v_3 - v_4) + (v_4 - v_1) = 0$. This is a nontrivial linear combination equaling zero, so they are linearly dependent. □

Week 2: Subsequences and Continuity

2.1. Let $L = \lim a_n = \lim b_n$. Given $\varepsilon > 0$, choose N such that $|a_n - L| < \varepsilon$ and $|b_n - L| < \varepsilon$ for $n \geq N$.
Since $a_n \leq c_n \leq b_n$, we have $a_n - L \leq c_n - L \leq b_n - L$.
If $c_n - L \geq 0$: $|c_n - L| = c_n - L \leq b_n - L \leq |b_n - L| < \varepsilon$.
If $c_n - L < 0$: $|c_n - L| = L - c_n \leq L - a_n = |a_n - L| < \varepsilon$.
Either way, $|c_n - L| < \varepsilon$ for $n \geq N$. Hence $\lim c_n = L$. □

2.3. Suppose f is not continuous at c. Then $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x$ with $|x - c| < \delta$ but $|f(x) - f(c)| \geq \varepsilon_0$.
Choose $\delta = \frac{1}{n}$. Then $\exists x_n$ with $|x_n - c| < \frac{1}{n}$ but $|f(x_n) - f(c)| \geq \varepsilon_0$.
This sequence $\{x_n\}$ satisfies $x_n \to c$, but $f(x_n) \not\to f(c)$, contradicting the hypothesis. Hence f is continuous at c. □

2.5. ($\Rightarrow$) If $U = V$, they have the same basis, so $\dim U = \dim V$.
($\Leftarrow$) Let $\{u_1, \ldots, u_n\}$ be a basis for U. Since $U \subseteq V$ and these are linearly independent vectors in V with $|\text{set}| = \dim V$, they form a basis for V. Hence $V = \text{span}\{u_1, \ldots, u_n\} = U$. □

2.7. $V = \{p \in \mathcal{P}_4 : p(1) = 0\}$ consists of polynomials with root at $x = 1$, so $p(x) = (x-1)q(x)$ where $\deg q \leq 3$.
Basis for V: $\{(x-1), x(x-1), x^2(x-1), x^3(x-1)\}$.
These span V since any $(x-1)(a + bx + cx^2 + dx^3)$ is a linear combination. They're independent since $(x-1)(c_1 + c_2 x + c_3 x^2 + c_4 x^3) = 0$ implies all $c_i = 0$.
Extend to $\mathcal{P}_4$: Add $\{1\}$. Check: $1 \notin V$ (since $1(1) = 1 \neq 0$). So basis for $\mathcal{P}_4$ is $\{1, (x-1), x(x-1), x^2(x-1), x^3(x-1)\}$. □

Week 3: Differentiation and Topology

3.1. Suppose $f : [a, b] \to \mathbb{R}$ is continuous but not uniformly continuous. Then $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x, y$ with $|x - y| < \delta$ but $|f(x) - f(y)| \geq \varepsilon_0$.
Take $\delta = 1/n$. Get sequences $\{x_n\}, \{y_n\}$ with $|x_n - y_n| < 1/n$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$.
By Bolzano-Weierstrass, $\{x_n\}$ has a convergent subsequence $x_{n_k} \to c \in [a, b]$. Since $|x_{n_k} - y_{n_k}| < 1/n_k \to 0$, we have $y_{n_k} \to c$ also.
By continuity: $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$, so $|f(x_{n_k}) - f(y_{n_k})| \to 0$. But this contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0$. Hence f is uniformly continuous. □

3.3. ($\Rightarrow$) If $a_n \to L$, then $\forall \varepsilon > 0$, $\exists N$ with $|a_n - L| < \varepsilon/2$ for $n \geq N$. For $m, n \geq N$:

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

($\Leftarrow$) First show $\{a_n\}$ is bounded: Take $\varepsilon = 1$. Then $\exists N$ with $|a_m - a_n| < 1$ for $m, n \geq N$. So $|a_m| \leq |a_N| + 1$ for $m \geq N$. Hence $\{a_n\}$ is bounded.
By BW, $\exists$ subsequence $a_{n_k} \to A$. Given $\varepsilon > 0$, choose K so $|a_{n_k} - A| < \varepsilon/2$ for $k \geq K$, and choose N so $|a_m - a_n| < \varepsilon/2$ for $m, n \geq N$. For $n \geq \max(N, n_K)$:

$$|a_n - A| \leq |a_n - a_{n_K}| + |a_{n_K} - A| < \varepsilon$$

Hence $a_n \to A$. □

3.5. Expanding:

$$\begin{aligned} \sum_{i,j} (a_i b_j - a_j b_i)^2 &= \sum_{i,j} (a_i^2 b_j^2 - 2 a_i b_j a_j b_i + a_j^2 b_i^2) \\ &= \sum_i a_i^2 \sum_j b_j^2 + \sum_j a_j^2 \sum_i b_i^2 - 2 \sum_i a_i b_i \sum_j a_j b_j \\ &= 2\|a\|^2 \|b\|^2 - 2(a \cdot b)^2 \end{aligned}$$

Hence $\frac{1}{2}\sum_{i,j}(a_ib_j - a_jb_i)^2 = \|a\|^2\|b\|^2 - (a \cdot b)^2 \geq 0$.
This gives $(a \cdot b)^2 \leq \|a\|^2\|b\|^2$, so $|a \cdot b| \leq \|a\|\|b\|$ (Cauchy-Schwarz). □

3.7. $(AB)^T_{ij} = (AB)_{ji} = \sum_k a_{jk}b_{ki}$.
$(B^TA^T)_{ij} = \sum_k (B^T)_{ik}(A^T)_{kj} = \sum_k b_{ki}a_{jk} = \sum_k a_{jk}b_{ki}$.
These are equal, so $(AB)^T = B^TA^T$. □

Week 4: Continuity and Metric Spaces

4.1. Define $h : [a,b] \to \mathbb{R}$ by $h(t) = (f(b) - f(a)) \cdot f(t)$. Then h is continuous on $[a,b]$ and differentiable on (a,b).
By MVT, $\exists c \in (a,b)$ with $h'(c) = \frac{h(b)-h(a)}{b-a}$.
Computing: $h(b) - h(a) = (f(b) - f(a)) \cdot f(b) - (f(b) - f(a)) \cdot f(a) = \|f(b) - f(a)\|^2$.
And $h'(c) = (f(b) - f(a)) \cdot f'(c)$.
So $\frac{\|f(b)-f(a)\|^2}{b-a} = (f(b) - f(a)) \cdot f'(c) \leq \|f(b) - f(a)\|\|f'(c)\|$ (Cauchy-Schwarz).
Hence $\|f(b) - f(a)\| \leq (b - a)\|f'(c)\|$. □

4.3. Let $c \in f^{-1}(U)$, so $f(c) \in U$. Since U is open, $\exists \varepsilon > 0$ with $B_\varepsilon(f(c)) \subseteq U$.
By continuity of f, $\exists \delta > 0$ with $d_X(x,c) < \delta \Rightarrow d_Y(f(x), f(c)) < \varepsilon$.
So $x \in B_\delta(c) \Rightarrow f(x) \in B_\varepsilon(f(c)) \subseteq U \Rightarrow x \in f^{-1}(U)$.
Hence $B_\delta(c) \subseteq f^{-1}(U)$, so $f^{-1}(U)$ is open. □

4.5. For $\text{rank}(AB) \leq \text{rank}(A)$: Each column of AB is a linear combination of columns of A (with coefficients from B). So $\text{col}(AB) \subseteq \text{col}(A)$, hence $\text{rank}(AB) \leq \text{rank}(A)$.
For $\text{rank}(AB) \leq \text{rank}(B)$: Each row of AB is a linear combination of rows of B. So $\text{row}(AB) \subseteq \text{row}(B)$, hence $\text{rank}(AB) \leq \text{rank}(B)$.
Therefore $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$. □

4.7. The projection of x onto $W = \text{span}\{v\}$ (where $\|v\| = 1$) is $(v \cdot x)v$.
In matrix form: $P_W(x) = v(v^Tx) = (vv^T)x$.
So the matrix is $P_W = vv^T$ (an $n \times n$ matrix).
The projection onto $W^\perp$ is $P_{W^\perp} = I - P_W = I - vv^T$. □

Week 5: Series and Higher-Dimensional Differentiation

5.1. By the ratio test hypothesis: $|a_n| \leq |a_N|\lambda^{n-N}$ for $n \geq N$.
So $\sum_{n=N}^\infty |a_n| \leq |a_N| \sum_{n=N}^\infty \lambda^{n-N} = |a_N| \cdot \frac{1}{1-\lambda}$ (geometric series with $\lambda < 1$).
Since $\sum_{n=1}^{N-1} |a_n|$ is finite, $\sum_{n=1}^\infty |a_n|$ converges. Hence $\sum a_n$ converges absolutely. □

5.3. Let $S_{2n} = \sum_{k=1}^{2n}(-1)^{k+1}a_k = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n})$.
Each parenthesis is ≥ 0 (since $\{a_n\}$ is decreasing), so S_{2n} is increasing.
Also $S_{2n} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots \leq a_1$, so S_{2n} is bounded.
Hence $S_{2n} \to L$ for some L.
Now $S_{2n+1} = S_{2n} + a_{2n+1} \to L + 0 = L$ (since $a_n \to 0$).
So $S_n \to L$. □

5.5. Since $\sum \|a_k\|$ converges, each component $\sum |a_{k,i}|$ converges (as $|a_{k,i}| \le \|a_k\|$).
For real series: if $\sum |b_k|$ converges, then $\sum b_k$ converges (define $b_k^+ = \max(b_k, 0)$, $b_k^- = \max(-b_k, 0)$; both are bounded by $|b_k|$).
So each component $\sum a_{k,i}$ converges. Hence $\sum a_k$ converges in $\mathbb{R}^n$. □

5.7. We compute det using row/column operations, tracking sign changes.
After reducing to upper triangular form (work omitted), we find det $= 15$.
Alternatively: The determinant equals $\prod_{1 \le i < j \le n}(x_j - x_i)$ for the Vandermonde matrix. □

Part II

Math 62CM: Differential Forms and Manifolds

Chapter 1

Dual Spaces and Topology

1.1 Dual Spaces

The highlight of this course is **Stokes' theorem**, which unifies Green's theorem, the divergence theorem, and the classical Stokes theorem into a single elegant statement. To develop this, we need multilinear algebra and the theory of differential forms.

Definition 1.1 · Dual Space

Let V be a real vector space. A function $f : V \to \mathbb{R}$ is called **linear** if $f(\lambda v) = \lambda f(v)$ and $f(v_1 + v_2) = f(v_1) + f(v_2)$ for all $\lambda \in \mathbb{R}$ and $v, v_1, v_2 \in V$.
The set of all linear functions on V forms a vector space called the **dual space**, denoted V^*.

Theorem 1.1 · Dimension of Dual Space

If $\dim(V) = n$ *is finite, then* $\dim(V^*) = n$.

Proof. Let $\{v_1, \ldots, v_n\}$ be a basis for V. Define $v_i^* : V \to \mathbb{R}$ by $v_i^*(v_j) = \delta_{ij}$ (the Kronecker delta). We claim $\{v_1^*, \ldots, v_n^*\}$ is a basis for V^*.
Linear independence: If $\sum a_i v_i^* = 0$, then applying to v_j gives $a_j = 0$ for all j.
Spanning: Given $f \in V^*$, let $a_i = f(v_i)$. Then $(f - \sum a_i v_i^*)(v_j) = f(v_j) - a_j = 0$, so $f = \sum a_i v_i^*$.
■

Definition 1.2 · Dual Basis

The basis $\{v_1^*, \ldots, v_n^*\}$ constructed above is called the **dual basis** to $\{v_1, \ldots, v_n\}$.

Remark. While V and V^* are isomorphic (they have the same dimension), there is no *canonical* isomorphism—it depends on a choice of basis. However, if V has an inner product, we get a canonical isomorphism $D : V \to V^*$ via $D(v) = \ell_v$ where $\ell_v(w) = \langle v, w \rangle$.

Theorem 1.2 · Double Dual

$(V^*)^*$ is canonically isomorphic to V (assuming $\dim V$ is finite).

Proof. Define $\iota : V \to (V^*)^*$ by $\iota(v)(f) = f(v)$ for $f \in V^*$. This is linear and injective (if $f(v) = 0$ for all f, then $v = 0$). Since $\dim V = \dim V^* = \dim(V^*)^*$, it's an isomorphism. ■

Definition 1.3 · Dual Map

If $A : V \to W$ is linear, the **dual map** $A^* : W^* \to V^*$ is defined by $(A^* f)(v) = f(Av)$.

In matrix form, if A is represented by matrix M, then A^* is represented by M^T.

1.2 Topology in Vector Spaces

Let V be a finite-dimensional Euclidean space (vector space with inner product).

Definition 1.4 · Open and Closed Sets

- The **open ball** of radius r centered at p is $B_r(p) = \{x \in V : \|x - p\| < r\}$.
- The **closed ball** is $\overline{B}_r(p) = \{x \in V : \|x - p\| \leq r\}$.
- The **sphere** is $S_r(p) = \{x \in V : \|x - p\| = r\}$.
- A set $U \subseteq V$ is **open** if for every $x \in U$, there exists $\delta > 0$ such that $B_\delta(x) \subseteq U$.
- A set $A \subseteq V$ is **closed** if $V \setminus A$ is open.

Theorem 1.3 · Properties of Open and Closed Sets

1. *Arbitrary unions of open sets are open.*
2. *Finite intersections of open sets are open.*
3. *Arbitrary intersections of closed sets are closed.*
4. *Finite unions of closed sets are closed.*

Definition 1.5 · Interior, Closure, Boundary

For $X \subseteq V$:

- The **interior** $\text{Int}(X)$ is the union of all open sets contained in X.
- The **closure** $\overline{X}$ is the intersection of all closed sets containing X.
- The **boundary** $\partial X = \overline{X} \setminus \text{Int}(X)$.

Definition 1.6 · Compact Sets

A set $A \subseteq V$ is **compact** if:

1. (Definition 1) A is closed and bounded.
2. (Definition 2) Every sequence $\{x_n\} \subseteq A$ has a subsequence converging to a point in A.
3. (Definition 3) Every open cover of A has a finite subcover.

These definitions are equivalent in $\mathbb{R}^n$ (Heine-Borel theorem).

1.3 Connected and Path-Connected Sets

Definition 1.7 · Connected

A set $A \subseteq V$ is **connected** if we cannot write $A = A_0 \cup A_1$ where A_0, A_1 are both non-empty, disjoint, and relatively open in A.

Definition 1.8 · Path-Connected

A set A is **path-connected** if for any $a, b \in A$, there exists a continuous path $\gamma : [0,1] \to A$ with $\gamma(0) = a$ and $\gamma(1) = b$.

Theorem 1.4 · Path-Connected Implies Connected

Proof. Suppose A is path-connected but not connected. Write $A = A_0 \cup A_1$ with A_0, A_1 open, disjoint, non-empty. Pick $a_0 \in A_0, a_1 \in A_1$ and a path $\gamma : [0,1] \to A$ connecting them. Then $f = \mathbf{1}_{A_1} \circ \gamma : [0,1] \to \{0,1\}$ is continuous with $f(0) = 0, f(1) = 1$. But this contradicts the Intermediate Value Theorem. ∎

Remark. The converse is false. The "topologist's sine curve" $\{(x, \sin(1/x)) : x > 0\} \cup \{(0,0)\}$ is connected but not path-connected.

Theorem 1.5 · $\mathbb{R}^n$ is Connected

$\mathbb{R}^n$ is connected (in fact, path-connected via linear paths $\gamma(t) = (1-t)x + ty$).

Definition 1.9 · Homeomorphism

$f : A \to B$ is a **homeomorphism** if f is continuous, bijective, and f^{-1} is continuous. We say A and B are **homeomorphic** ("topologically the same").

Theorem 1.6 · Continuous Image of Connected

Continuous images of connected sets are connected.

Exercises

Exercise 1.1. Prove the converse: given any basis $\ell_1, \ldots, \ell_n$ of V^*, construct a dual basis $w_1, \ldots, w_n$ of V so that $\ell_i(w_j) = \delta_{ij}$.

Exercise 1.2. Let V be a vector space with inner product $\langle \cdot, \cdot \rangle$. Prove that $\mathcal{D} : V \to V^*$, $\mathcal{D}(v) = \ell_v$ where $\ell_v(x) = \langle v, x \rangle$, is an isomorphism. Show that $\mathcal{D} = i_S$ for any orthonormal basis S.

Exercise 1.3. Show that both maps i_S and $\mathcal{D}$ are injective in the infinite-dimensional case as well.

Exercise 1.4. Prove that the definition of open and closed sets is independent of the choice of inner product on a finite-dimensional vector space.

Exercise 1.5. Show that the closure of a connected set is connected.

Exercise 1.6. Prove that the closure of the graph of $\sin(1/x)$, $x > 0$, is connected but not path-connected.

Exercise 1.7.

1. Prove that $[0,1]$, $[0,1)$, and $(0,1)$ are pairwise not homeomorphic.
2. Prove that $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}$ if $n > 1$.

Chapter 2

Differential Forms

2.1 Vector Fields and 1-Forms

Definition 2.1 · Vector Field

A **vector field** on an open set $U \subseteq V$ is an assignment $x \mapsto v_x \in V_x$ (where V_x is a copy of V at x).

Definition 2.2 · Differential 1-Form

A **differential 1-form** α on U is an assignment $x \mapsto \alpha_x \in V_x^*$ varying smoothly.

In coordinates $(x_1, \ldots, x_n)$, a 1-form is $\alpha = \sum a_i(x) dx_i$ where dx_i are the coordinate differentials (dual to the standard basis).

Definition 2.3 · Differential of a Function

For $f : U \to \mathbb{R}$ differentiable, the **differential** df is a 1-form defined by:

$$df_x(v) = \lim_{t \to 0} \frac{f(x + tv) - f(x)}{t} = D_v f(x)$$

In coordinates: $df = \sum \frac{\partial f}{\partial x_i} dx_i$.

Definition 2.4 · Exact and Closed Forms

A 1-form α is:

- **Exact** if $\alpha = df$ for some function f (called a **primitive**).
- **Closed** if $\frac{\partial a_i}{\partial x_j} = \frac{\partial a_j}{\partial x_i}$ for all i, j.

Theorem 2.1 · Exact Implies Closed

Exact $\Rightarrow$ Closed.

2.2 Pullback

Definition 2.5 · Pullback

Let $f : U \to U'$ be differentiable and α a 1-form on U'. The **pullback** $f^*\alpha$ is a 1-form on U defined by:

$$(f^*\alpha)_x(v) = \alpha_{f(x)}((d_x f)(v))$$

Theorem 2.2 · Properties of Pullback

1. $f^*(\alpha + \beta) = f^*\alpha + f^*\beta$
2. $f^*(\phi\alpha) = (\phi \circ f) f^*\alpha$ *for functions* ϕ
3. $f^*(dg) = d(g \circ f)$
4. $(g \circ f)^* = f^* \circ g^*$

Example: Let $f(r, \theta) = (r\cos\theta, r\sin\theta)$ (polar coordinates) and $\alpha = xdy - ydx$ on $\mathbb{R}^2$. Then:

$$f^*\alpha = r\cos\theta \cdot d(r\sin\theta) - r\sin\theta \cdot d(r\cos\theta) = r^2 d\theta$$

2.3 Integration on Curves

Definition 2.6 · Line Integral

Let $\gamma : [a, b] \to U$ be a smooth path and α a 1-form on U. The **line integral** is:

$$\int_\gamma \alpha = \int_a^b \gamma^*\alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))dt$$

Theorem 2.3 · Independence of Parametrization

The line integral is independent of parametrization (up to sign for orientation reversal).

Theorem 2.4 · Fundamental Theorem for Exact Forms

If $\alpha = df$ *is exact and* γ *is a path from* A *to* B*:*

$$\int_\gamma df = f(B) - f(A)$$

In particular, $\oint_\gamma df = 0$ *for any closed loop.*

2.4 Poincaré Lemma

Definition 2.7 · Star-Shaped

A set U is **star-shaped** with respect to $a \in U$ if for every $x \in U$ and $t \in [0,1]$, $(1-t)a + tx \in U$.

Theorem 2.5 · Poincaré Lemma

On a star-shaped open set U, every closed 1-form is exact.

Proof. Let $\alpha = \sum a_i(x) dx_i$ be closed. Define:

$$f(x) = \int_0^1 \sum a_i(tx) x_i \, dt$$

Then $df = \alpha$. ∎

Definition 2.8 · Simply Connected

U is **simply connected** if it is path-connected and every two paths connecting the same endpoints are homotopic.

Theorem 2.6 · Simply Connected Poincaré

On a simply connected open set, every closed form is exact.

Example: The form $\alpha = \frac{-ydx+xdy}{x^2+y^2}$ on $\mathbb{R}^2 \setminus \{0\}$ is closed but not exact (since $\oint_{S^1} \alpha = 2\pi \neq 0$). This shows $\mathbb{R}^2 \setminus \{0\}$ is not simply connected.

Exercises

Exercise 2.1. Let $f : \mathbb{R}^2 \to \mathbb{R}^3$, $f(x,y) = (e^x - 2y, x + y^2, \sin(y) + 1)$ and $\alpha = z\,dx - 3\,dy + e^y\,dz$. Compute $f^*\alpha$.

Exercise 2.2. In $\mathbb{R}^4$ with coordinates (x,y,z,u), consider the plane field defined by $dz - y\,dx = 0$ and $dy - u\,dx = 0$. For a curve Γ given by $y = f(x)$, $z = g(x)$, $u = h(x)$, find necessary and sufficient conditions for Γ to be tangent to this plane field.

Exercise 2.3. In $\mathbb{R}^3$, find a curve Γ whose projection to the (x,y)-plane is $x = y^2$, tangent to $dz - y\,dx = 0$, and containing the origin.

Exercise 2.4. Find a primitive in $\mathbb{R}^2$ of the exact 1-form:

$$\alpha = e^x(e^y(x - y + 2) + y)dx + e^x(e^y(x - y) + 1)dy$$

Exercise 2.5. Prove that $\mathbb{R}^3 \setminus \{y = x^2, z = 0\}$ is not simply connected.

Exercise 2.6. Show that any star-shaped open set is simply connected.

Chapter 3

Multilinear Algebra

3.1 Bilinear and Multilinear Forms

Definition 3.1 · Multilinear Form

A function $f : V \times \cdots \times V \to \mathbb{R}$ (k copies) is **k-linear** (or multilinear) if it is linear in each argument when the others are fixed.

Definition 3.2 · Bilinear Form

A **bilinear form** $f : V \times V \to \mathbb{R}$ is **symmetric** if $f(x, y) = f(y, x)$ and **skew-symmetric** if $f(x, y) = -f(y, x)$.

Definition 3.3 · Quadratic Form

A **quadratic form** is $Q(x) = f(x, x)$ for some symmetric bilinear f.

Theorem 3.1 · Polarization

Given a quadratic form Q, there is a unique symmetric bilinear form f with $Q(x) = f(x, x)$:

$$f(x, y) = \frac{1}{2}(Q(x + y) - Q(x) - Q(y))$$

Definition 3.4 · Index and Rank

For a quadratic form Q written as $Q = \sum \lambda_j x_j^2$:

- The number of positive λ_j is the **positive index**.
- The number of negative λ_j is the **negative index**.
- The rank is the total number of nonzero λ_j.

Theorem 3.2 · Sylvester's Law of Inertia

The positive index, negative index, and nullity of a quadratic form are independent of the choice of basis (i.e., the presentation as a sum of squares).

3.2 Exterior (Wedge) Product

Definition 3.5 · Skew-Symmetric k-Linear Form

$f : V^k \to \mathbb{R}$ is **skew-symmetric** if swapping any two arguments negates the value:

$$f(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -f(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k)$$

The space of skew-symmetric k-linear forms on V is denoted $\Lambda^k(V^*)$ and has dimension $\binom{n}{k}$ where $n = \dim V$.

Definition 3.6 · Wedge Product

For $\varphi \in \Lambda^k(V^*)$ and $\psi \in \Lambda^\ell(V^*)$, the **wedge product** $\varphi \wedge \psi \in \Lambda^{k+\ell}(V^*)$ is:

$$(\varphi \wedge \psi)(v_1, \ldots, v_{k+\ell}) = \frac{1}{k!\ell!} \sum_{\sigma \in S_{k+\ell}} \operatorname{sign}(\sigma) \varphi(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \psi(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+\ell)})$$

Theorem 3.3 · Properties of Wedge Product

1. $\varphi \wedge \psi = (-1)^{k\ell} \psi \wedge \varphi$
2. $(\varphi \wedge \psi) \wedge \chi = \varphi \wedge (\psi \wedge \chi)$
3. $\ell_1 \wedge \cdots \wedge \ell_k = 0$ *if and only if* $\ell_1, \ldots, \ell_k$ *are linearly dependent*

Theorem 3.4 · Basis for $\Lambda^k(V^*)$

If $x_1, \ldots, x_n$ *is a basis for* V^*, *then* $\{x_{i_1} \wedge \cdots \wedge x_{i_k} : i_1 < \cdots < i_k\}$ *is a basis for* $\Lambda^k(V^*)$.

3.3 Determinant as a Volume Form

Theorem 3.5 · Determinant via Exterior Power

For a linear map $A : V \to V$, *the induced map* $A^* : \Lambda^n(V^*) \to \Lambda^n(V^*)$ *is multiplication by* $\det(A)$.

This gives an intrinsic definition of the determinant: it is the scalar by which A scales the "top" exterior power.

Definition 3.7 · Orientation

An **orientation** on V is an equivalence class of ordered bases, where two bases are equivalent if the change-of-basis matrix has positive determinant.

Definition 3.8 · Volume

For linearly independent $v_1, \ldots, v_k \in V$ with inner product:

$$\mathrm{Vol}(P(v_1, \ldots, v_k)) = |\det(U)|$$

where U is the matrix with columns v_i in an orthonormal basis. Equivalently, $\mathrm{Vol} = \sqrt{\det(v_i \cdot v_j)}$ (the Gram determinant).

3.4 Tensor Products

Definition 3.9 · Tensor Product

The **tensor product** $V \otimes W$ is defined by the universal property: there is a bilinear map $V \times W \to V \otimes W$, $(v, w) \mapsto v \otimes w$, and any bilinear map $V \times W \to Z$ factors uniquely through $V \otimes W$.

Theorem 3.6 · Dimension of Tensor Product

$\dim(V \otimes W) = \dim(V) \cdot \dim(W)$. *If $\{v_i\}$ is a basis for V and $\{w_j\}$ is a basis for W, then $\{v_i \otimes w_j\}$ is a basis for $V \otimes W$.*

Theorem 3.7 · Tensor-Hom Adjunction

$V \otimes W \cong Hom(V^*, W)$ via $F(v \otimes w)(g) = g(v)w$.

Exercises

Exercise 3.1. Let M be the space of $n \times n$ real matrices. For the bilinear form $B(X, Y) = \mathrm{Tr}(XY)$, show B is symmetric and compute the rank and index of its associated quadratic form.

Exercise 3.2.

1. Prove that any exterior 2-form on an odd-dimensional space is degenerate.
2. Show that the rank of any skew-symmetric bilinear form is even.

Exercise 3.3. Let ω be an exterior 2-form on a $2n$-dimensional space V.

1. Prove ω is non-degenerate iff $\omega^{\wedge n} \neq 0$.

2. Prove there exists a basis where ω has the standard symplectic matrix form.

Exercise 3.4. Show that $S^2(V)$ (the symmetric square) is generated by tensors of the form $v \otimes v$.

Exercise 3.5. Compute the 3-dimensional volume of each face of the parallelepiped $P(v_1, v_2, v_3, v_4) \subset \mathbb{R}^4$ where $v_1 = (1,1,1,1)$, $v_2 = (1,-1,1,1)$, $v_3 = (1,1,-1,1)$, $v_4 = (1,1,1,-1)$.

Chapter 4

Differential Forms on Open Sets

4.1 k-Forms and the Exterior Derivative

Definition 4.1 · Differential k-Form

A **differential k-form** on an open set $U \subseteq \mathbb{R}^n$ is:

$$\omega = \sum_{i_1 < \cdots < i_k} f_{i_1 \ldots i_k}(x)\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

where the $f_{i_1 \ldots i_k}$ are smooth functions on U.

The space of k-forms on U is denoted $\Omega^k(U)$.

Definition 4.2 · Exterior Derivative

The **exterior derivative** $d : \Omega^k(U) \to \Omega^{k+1}(U)$ is defined by:

$$d\omega = \sum_{i_1 < \cdots < i_k} df_{i_1 \ldots i_k} \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

Theorem 4.1 · Properties of d

1. *$d(d\omega) = 0$ for all ω*
2. *$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ (graded Leibniz rule)*
3. *d commutes with pullback: $d(f^*\omega) = f^*(d\omega)$*

Definition 4.3 · Closed and Exact Forms

A k-form ω is:

- **Closed** if $d\omega = 0$
- **Exact** if $\omega = d\eta$ for some $(k-1)$-form η

Since $d^2 = 0$: Exact $\Rightarrow$ Closed.

4.2 Vector Calculus Operations

In $\mathbb{R}^3$, the exterior derivative captures all classical operations:

- **Gradient**: $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz \leftrightarrow \nabla f$
- **Curl**: $d(P\,dx + Q\,dy + R\,dz) = (\text{curl components}) \leftrightarrow \nabla \times \mathbf{F}$
- **Divergence**: $d(P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy) = (\nabla \cdot \mathbf{F})dx \wedge dy \wedge dz$

The identities $\nabla \times (\nabla f) = 0$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ follow from $d^2 = 0$.

4.3 The Hodge Star Operator

Definition 4.4 · Hodge Star

On an oriented Euclidean space V of dimension n, the **Hodge star** $* : \Lambda^k(V^*) \to \Lambda^{n-k}(V^*)$ satisfies:

$$\alpha \wedge (*\beta) = \langle \alpha, \beta \rangle \,\mathrm{vol}$$

where vol is the volume form.

Example: In $\mathbb{R}^3$: $*dx = dy \wedge dz$, $*dy = dz \wedge dx$, $*dz = dx \wedge dy$, $*(dx \wedge dy) = dz$, etc.

4.4 Change of Variables

Theorem 4.2 · Change of Variables Formula

Let $\varphi : A \to B$ be a C^1 diffeomorphism between measurable sets. Then for integrable f:

$$\int_B f(y)\,dy = \int_A (f \circ \varphi)(x)|\det(d\varphi_x)|\,dx$$

Theorem 4.3 · Fubini's Theorem

For $f : \mathbb{R}^{n+m} \to \mathbb{R}$ integrable:

$$\int_{\mathbb{R}^{n+m}} f = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^m} f(x,y)\,dy \right) dx$$

Exercises

Exercise 4.1. Let $h : \mathbb{R} \to \mathbb{R}$ be smooth with $\lim_{x\to\pm\infty} h(x) = \pm\infty$ and $h' > 0$. Prove h is a diffeomorphism.

Exercise 4.2. Is $x \mapsto x^3$ a diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$?

Exercise 4.3. Prove that $\phi(x) = \begin{cases} e^{-1/x^2} & x > 0 \\ 0 & x \leq 0 \end{cases}$ is $C^\infty(\mathbb{R})$.

Exercise 4.4. Compute $\int_\gamma f(x^2 + y^2 + z^2)(x\,dx + y\,dy + z\,dz)$ where $\gamma(t) = (5\sin^2(20t), 2 - 2\cos(t/3), \sin(t)/2)$ for $t \in [0, \pi]$ and $\int_0^1 f(t)\,dt = 1$.

Chapter 5

Manifolds

5.1 Submanifolds

Definition 5.1 · Submanifold

A subset $M \subseteq V$ is a k**-dimensional submanifold** if for every $p \in M$, there exists a neighborhood U_p of p in V and a diffeomorphism $\varphi : U_p \to \mathbb{R}^n$ such that:

$$\varphi(M \cap U_p) = \{(y_1, \dots, y_n) : y_{k+1} = \cdots = y_n = 0\}$$

Equivalently:

- **As a graph**: Locally $M = \{(x, f(x)) : x \in U \subseteq \mathbb{R}^k\}$ for some smooth f.
- **As a level set**: Locally $M = \{F_1 = \cdots = F_{n-k} = 0\}$ where $dF_1, \dots, dF_{n-k}$ are linearly independent.
- **As an image**: $M = \mathrm{Im}(\varphi)$ for an embedding $\varphi : U \to V$.

Example: $S^{n-1} = \{x \in \mathbb{R}^n : \|x\|^2 = 1\}$ is a manifold (level set of $F(x) = \|x\|^2 - 1$).

5.2 Tangent Space

Definition 5.2 · Tangent Space

For M a manifold and $p \in M$:

- If $M = \mathrm{Im}(\varphi)$, then $T_pM = \mathrm{Im}(d_{\varphi^{-1}(p)}\varphi)$.
- If $M = \{F = 0\}$, then $T_pM = \ker(d_pF)$.

The tangent space is independent of the parametrization/presentation.

Definition 5.3 · Tangent Bundle

The **tangent bundle** is $TM = \{(p,v) : p \in M, v \in T_pM\} \subseteq V \times V$.

TM is a $2k$-dimensional manifold.

5.3 Manifolds with Boundary

Definition 5.4 · Manifold with Boundary

A k-dimensional **manifold with boundary** M is locally diffeomorphic to either $\mathbb{R}^k$ or $\mathbb{R}^k_+ = \{x_k \geq 0\}$.
The **boundary** ∂M consists of points mapping to $\{x_k = 0\}$; it is a $(k-1)$-dimensional manifold.

> Example: $D^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ is a manifold with boundary $\partial D^n = S^{n-1}$.

5.4 Orientation and Integration

Definition 5.5 · Orientation

An **orientation** on a manifold M is a continuous choice of orientation on each tangent space T_pM.

Definition 5.6 · Integration of k-Forms

For M an oriented k-dimensional manifold and ω a k-form with compact support:

$$\int_M \omega$$

is defined using partitions of unity and local parametrizations.

Exercises

Exercise 5.1. Let $SO(3) \subset \mathcal{M}_{3,3}$ be the group of orthogonal matrices with determinant $+1$. Let $UT(S^2) = \{(x,y) \in \mathbb{R}^3 \times \mathbb{R}^3 : \|x\| = \|y\| = 1, x \cdot y = 0\}$.

1. Show that $SO(3)$ and $UT(S^2)$ are submanifolds of $\mathbb{R}^9$ and $\mathbb{R}^6$ respectively.
2. Prove they are diffeomorphic.

Exercise 5.2. Consider the torus $T = \{z^2 + (r-2)^2 = 1\}$ in cylindrical coordinates and $T' = S^1 \times S^1 \subset \mathbb{R}^4$.

1. Prove T is a submanifold of $\mathbb{R}^3$.

2. Prove T and T' are diffeomorphic.

3. Compute the surface areas of T and T'.

Exercise 5.3. Let $f : \mathbb{R}^n \to \mathbb{R}^{n(n+1)/2}$ map x to all products $x_i x_j$. Let $M = f(S^{n-1})$ (the real projective space).

1. Describe $f^{-1}(y) \cap S^{n-1}$ for $y \in M$.

2. Prove M is an $(n-1)$-dimensional manifold.

Chapter 6

Stokes' Theorem and Cohomology

6.1 Stokes' Theorem

Theorem 6.1 · Stokes' Theorem

Let M be an oriented k-dimensional manifold with boundary, and let ω be a $(k-1)$-form with compact support. Then:

$$\int_M d\omega = \int_{\partial M} \omega$$

where ∂M has the induced orientation.

Special cases:

- $k = 1$ (curves): $\int_a^b df = f(b) - f(a)$ (Fundamental Theorem of Calculus)
- $k = 2$ in $\mathbb{R}^2$ (Green's Theorem): $\int_D (Q_x - P_y)\, dA = \oint_{\partial D} P\, dx + Q\, dy$
- $k = 2$ surfaces in $\mathbb{R}^3$ (Classical Stokes): $\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$
- $k = 3$ in $\mathbb{R}^3$ (Divergence Theorem): $\int_V \nabla \cdot \mathbf{F}\, dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}$

6.2 De Rham Cohomology

Definition 6.1 · De Rham Cohomology

The k**th de Rham cohomology** of a manifold M is:

$$H^k(M; \mathbb{R}) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}} = \frac{\ker(d : \Omega^k \to \Omega^{k+1})}{\operatorname{Im}(d : \Omega^{k-1} \to \Omega^k)}$$

Theorem 6.2 · Poincaré Lemma

If U is contractible, then $H^k(U; \mathbb{R}) = 0$ for $k \geq 1$.

Theorem 6.3 · Homotopy Invariance

If $f, g : M \to N$ are homotopic, then $f^* = g^* : H^k(N) \to H^k(M)$.

Homotopy equivalent spaces have isomorphic cohomology.

Example: $H^1(S^1; \mathbb{R}) = \mathbb{R}$, generated by $d\theta$. This shows S^1 is not contractible.

6.3 Applications

Theorem 6.4 · No Retraction

There is no continuous retraction from D^n onto S^{n-1}.

Proof. If $r : D^n \to S^{n-1}$ is a retraction with $r|_{S^{n-1}} = \mathrm{id}$, let ω be a closed non-exact form on S^{n-1}. Then $r^*\omega$ is closed on D^n, hence exact: $r^*\omega = d\beta$. But $\int_{S^{n-1}} \omega = \int_{S^{n-1}} r^*\omega = \int_{D^n} d(r^*\omega) = 0$ by Stokes, contradicting ω being non-exact. ■

Theorem 6.5 · Brouwer Fixed Point Theorem

Every continuous map $f : D^n \to D^n$ has a fixed point.

Proof. If $f(x) \neq x$ for all x, define $r(x)$ as the intersection of the ray from $f(x)$ through x with S^{n-1}. This gives a retraction, contradicting the previous theorem. ■

Exercises

Exercise 6.1. Compute $\int_C (x^2 - y)dx + (x - y^2)dy$ where C is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ oriented counter-clockwise.

Exercise 6.2. Let $f : U \to \mathbb{R}$ be harmonic ($\Delta f = 0$) on an open set $U \subseteq \mathbb{R}^2$. Prove that $f(x, y)$ equals the average of f on any circle around (x, y) contained in U.

Exercise 6.3. For $\mathbf{v} = (x^2 + y^2 + z^2)(x\partial_x + y\partial_y + z\partial_z)$, compute $\mathrm{Work}_\Gamma(\mathbf{v})$ where Γ connects $A = (a_1, a_2, a_3)$ to $B = (b_1, b_2, b_3)$.

Exercise 6.4. Compute $\int_S \frac{dy \wedge dz}{x} + \frac{dz \wedge dx}{y} + \frac{dx \wedge dy}{z}$ where S is the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Exercise 6.5. Use Stokes' theorem to compute $\int_C (y + z)dx + (z + x)dy + (x + y)dz$ where C is the curve $x = \sin^2 t$, $y = 2 \sin t \cos t$, $z = \cos^2 t$ for $t \in [0, \pi]$.

Chapter 7

Advanced Topics: Degree Theory and Linking Numbers

7.1 Degree of a Map

The degree is a powerful topological invariant that counts (with signs) how many times a map "wraps" one manifold around another.

Definition 7.1 · Degree

Let $f : M \to N$ be a smooth map between compact, connected, oriented n-manifolds without boundary. The **degree** of f is:

$$\deg(f) = \int_M f^*\omega \bigg/ \int_N \omega$$

where ω is any n-form on N with $\int_N \omega \neq 0$.

Theorem 7.1 · Degree is an Integer

The degree is always an integer.

Theorem 7.2 · Local Formula for Degree

If $b \in N$ is a regular value of f, then:

$$\deg(f) = \sum_{a \in f^{-1}(b)} sign(\det(d_a f))$$

This is the signed count of preimages.

Theorem 7.3 · Homotopy Invariance

If $f_0, f_1 : M \to N$ are homotopic, then $\deg(f_0) = \deg(f_1)$.

Example: For $f : S^1 \to S^1$ given by $f(e^{i\theta}) = e^{in\theta}$, we have $\deg(f) = n$. This is the "winding number."

Example: Let A be a 3×3 matrix with $\det(A) \neq 0$. Define $F_A : S^2 \to S^2$ by $F_A(x) = \frac{Ax}{\|Ax\|}$. Since A is invertible, F_A is a diffeomorphism onto its image. The degree depends on orientation: $\deg(F_A) = \text{sign}(\det(A))$.

7.2 Winding and Linking Numbers

Definition 7.2 · Winding Number

Let $\gamma : [0, 2\pi] \to \mathbb{R}^2 \setminus \{p\}$ be a closed loop. The **winding number** of γ around p is:

$$w(\gamma, p) = \frac{1}{2\pi} \oint_\gamma \frac{-(y - p_2)dx + (x - p_1)dy}{(x - p_1)^2 + (y - p_2)^2}$$

Theorem 7.4 · Winding as Degree

The winding number equals the degree of the map $\gamma/|\gamma - p| : S^1 \to S^1$.

Definition 7.3 · Linking Number

Let $\phi, \psi : S^1 \to \mathbb{R}^3$ be two disjoint closed curves. The **linking number** $\text{lk}(\phi, \psi)$ measures how many times one curve winds around the other.
If D is a disk bounded by ϕ, then:

$$\text{lk}(\phi, \psi) = \sum_{\psi(t) \in D} \text{sign}(\psi'(t) \cdot n)$$

where n is the normal to D.

Theorem 7.5 · Gauss Linking Integral

$$lk(\phi, \psi) = \frac{1}{4\pi} \oint_\phi \oint_\psi \frac{(\phi - \psi) \cdot (d\phi \times d\psi)}{|\phi - \psi|^3}$$

Example: The Hopf link consists of two circles in $\mathbb{R}^3$ that are linked once. Their linking number is ± 1 (depending on orientations).

7.3 Higher Connectivity

Definition 7.4 · k-Connected

A space A is **k-connected** if any continuous map $f : S^m \to A$ for $m \leq k$ is homotopic to a constant map.
Equivalently, $H^i(A; \mathbb{R}) = 0$ for all $i \leq k$.

Theorem 7.6 · Connectivity of Spheres

$\mathbb{R}^{n+1} \setminus \{0\}$ (equivalently, S^n) is $(n-1)$-connected but not n-connected.

Sketch. $\mathbb{R}^{n+1} \setminus \{0\}$ is homotopy equivalent to S^n via $x \mapsto x/\|x\|$.
For $m < n$: any map $S^m \to S^n$ can be deformed to miss a point, hence is homotopic to a constant (the target minus a point is contractible).
For $m = n$: the identity map $S^n \to S^n$ has degree 1, so it's not homotopic to a constant (which has degree 0). ∎

Theorem 7.7 · Cohomology of T^2

For the torus $T^2 = S^1 \times S^1$ with coordinates ϕ_1, ϕ_2:

- $H^0(T^2) = \mathbb{R}$, *generated by the constant 1*
- $H^1(T^2) = \mathbb{R}^2$, *generated by $[d\phi_1]$ and $[d\phi_2]$*
- $H^2(T^2) = \mathbb{R}$, *generated by $[d\phi_1 \wedge d\phi_2]$*

7.4 Lie Derivatives (Brief Overview)

The Lie derivative measures how a tensor field changes along the flow of a vector field.

Definition 7.5 · Flow of a Vector Field

Let X be a vector field on $U \subseteq \mathbb{R}^n$. The **flow** of X is the family of diffeomorphisms $\phi_t : U \to U$ satisfying:

$$\frac{d}{dt}\phi_t(p) = X(\phi_t(p)), \quad \phi_0 = \mathrm{id}$$

Intuitively, $\phi_t(p)$ follows the integral curve of X starting at p for time t.

Definition 7.6 · Lie Derivative of a Function

For a smooth function f and vector field X:

$$\mathcal{L}_X f = \frac{d}{dt}\Big|_{t=0}(\phi_t^* f) = \frac{d}{dt}\Big|_{t=0} f \circ \phi_t = X(f) = df(X)$$

This is just the directional derivative of f along X.

Definition 7.7 · Lie Derivative of a Differential Form

For a k-form ω and vector field X:

$$\mathcal{L}_X\omega = \frac{d}{dt}\Big|_{t=0}\phi_t^*\omega$$

This measures the "rate of change" of ω as we flow along X.

Theorem 7.8 · Cartan's Magic Formula

For a vector field X and k-form ω:

$$\mathcal{L}_X\omega = d(\iota_X\omega) + \iota_X(d\omega)$$

*where ι_X is the **interior product** (contraction): $(\iota_X\omega)(v_1, \ldots, v_{k-1}) = \omega(X, v_1, \ldots, v_{k-1})$.*

Remark. Cartan's formula is incredibly useful because it expresses the Lie derivative purely in terms of d and ι_X, without needing to compute the flow explicitly.

Definition 7.8 · Lie Bracket

For vector fields X, Y, the **Lie bracket** $[X, Y]$ is the vector field defined by:

$$[X, Y](f) = X(Y(f)) - Y(X(f))$$

This measures the "failure of commutativity" of flows: if $[X, Y] = 0$, the flows of X and Y commute.

Theorem 7.9 · Properties of Lie Derivative

1. *$\mathcal{L}_X$ is a derivation: $\mathcal{L}_X(\omega \wedge \eta) = (\mathcal{L}_X\omega) \wedge \eta + \omega \wedge (\mathcal{L}_X\eta)$*
2. *$\mathcal{L}_X$ commutes with d: $\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega)$*
3. *$[\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_{[X,Y]}$ (Jacobi identity)*

Exercises

Exercise 7.1. Prove that $\mathbb{R}^2 \setminus \{(1,0), (-1,0)\}$ is not diffeomorphic to $\mathbb{R}^2 \setminus \{(0,0)\}$.

Exercise 7.2. Show that there are no degree 1 maps $f : S^2 \to T^2$.

Exercise 7.3. Let A be a 3×3 matrix with $\det(A) \neq 0$. Compute $\deg(F_A)$ where $F_A(x) = Ax/\|Ax\|$ for $x \in S^2$.

Exercise 7.4. Let $\gamma = (\gamma_1, \gamma_2) : [0, 2\pi] \to \mathbb{R}^2 \setminus \{(-1,0), (1,0)\}$ be a loop with winding number 2 around $(-1,0)$ and -3 around $(1,0)$. Compute $\mathrm{lk}(\phi, \psi)$ where

$\phi(s) = (\cos s, \sin s, 0)$ and $\psi(t) = (\gamma_1(t), 0, \gamma_2(t))$.

Solutions to Math 62 Exercises

Chapter 1: Dual Spaces and Topology

1.1. Given a basis $\{\ell_1, \dots, \ell_n\}$ of V^*, we want to construct vectors $w_1, \dots, w_n \in V$ with $\ell_i(w_j) = \delta_{ij}$.
Define $\Phi : V \to \mathbb{R}^n$ by $\Phi(v) = (\ell_1(v), \dots, \ell_n(v))$. We claim Φ is an isomorphism.
Injective: If $\Phi(v) = 0$, then $\ell_i(v) = 0$ for all i. Since $\{\ell_i\}$ is a basis for V^*, this means $f(v) = 0$ for all $f \in V^*$, so $v = 0$.
Surjective: $\dim V = \dim V^* = n = \dim \mathbb{R}^n$, and Φ is injective, so Φ is bijective.
Now let $e_1, \dots, e_n$ be the standard basis of $\mathbb{R}^n$. Define $w_j = \Phi^{-1}(e_j)$. Then:

$$\ell_i(w_j) = (\Phi(w_j))_i = (e_j)_i = \delta_{ij}$$

□

1.3. **For $\mathcal{D}$:** We need $\ker \mathcal{D} = \{0\}$. If $\mathcal{D}(v) = 0$, then $\langle v, x \rangle = 0$ for all $x \in V$. Taking $x = v$ gives $\|v\|^2 = 0$, so $v = 0$. Hence $\mathcal{D}$ is injective.
For i_S: Suppose $i_S(v) = 0$, meaning $v^*(x) = 0$ for all x. Write $v = \sum a_k v_k$ in the basis. If $v \neq 0$, some $a_k \neq 0$, and $v^*(v_k) = a_k \neq 0$, contradiction. So $v = 0$. □

1.5. Let A be connected and suppose $\overline{A}$ is disconnected: $\overline{A} = U \cup V$ with U, V open in $\overline{A}$, disjoint, and non-empty.
Since $A \subseteq \overline{A}$, we have $A = (A \cap U) \cup (A \cap V)$. These are relatively open in A and disjoint.
If both $A \cap U$ and $A \cap V$ are non-empty, then A is disconnected, contradiction.
So WLOG, $A \subseteq U$. Then $\overline{A} \subseteq \overline{U}$. But U is closed in $\overline{A}$ (complement of open V), so $\overline{U} \cap \overline{A} = U$. Hence $\overline{A} \subseteq U$, meaning $V = \emptyset$, contradiction. □

1.7. **(a)** $[0, 1]$ vs $[0, 1)$: Remove two points from $[0, 1]$ to disconnect it (remove interior points). But removing any two points from $[0, 1)$: if we remove 0 and another point a, we get $(0, a) \cup (a, 1)$ which is disconnected. If we remove two interior points, we get three components. The key is that $[0, 1]$ has exactly 2 "special" endpoints that don't disconnect when removed, while $[0, 1)$ has only 1.
$[0, 1)$ vs $(0, 1)$: Removing one point from $[0, 1)$: if we remove 0, it stays connected. If we remove an interior point, it disconnects. For $(0, 1)$, removing *any* point disconnects it. So they're not homeomorphic.

(b) Remove a point from $\mathbb{R}^n$ ($n > 1$): the result $\mathbb{R}^n \setminus \{p\}$ is still path-connected (go around the point). But removing a point from $\mathbb{R}$ gives two disconnected half-lines. A homeomorphism would preserve connectedness of point-removals. □

Chapter 2: Differential Forms

2.1. $f(x,y) = (e^x - 2y, x + y^2, \sin y + 1)$, $\alpha = z\,dx - 3\,dy + e^y\,dz$.
Using $f^*(dg) = d(g \circ f)$ and linearity:

$$\begin{aligned} f^*\alpha &= (\sin y + 1)d(e^x - 2y) - 3d(x + y^2) + e^{x+y^2} d(\sin y + 1) \\ &= (\sin y + 1)(e^x dx - 2dy) - 3(dx + 2y\,dy) + e^{x+y^2} \cos y\,dy \\ &= (e^x \sin y + e^x - 3)dx + (-2\sin y - 2 - 6y + e^{x+y^2} \cos y)dy \end{aligned}$$

□

2.3. The projection gives $x = y^2$, so parameterize as $(y^2, y, h(y))$. The plane field condition $dz - y\,dx = 0$ becomes:

$$h'(y)\,dy - y \cdot 2y\,dy = 0 \implies h'(y) = 2y^2 \implies h(y) = \frac{2}{3}y^3 + C$$

With $(0,0,0) \in \Gamma$: $h(0) = 0$, so $C = 0$.
The curve is $\Gamma(t) = (t^2, t, \frac{2}{3}t^3)$. □

2.5. Consider $\alpha = \frac{-z\,dy + y\,dz}{y^2+z^2}$ on $\mathbb{R}^3 \setminus P$.
Let $\Gamma = \{x = 0, y^2 + z^2 = 1\}$ parameterized by $\gamma(t) = (0, \cos t, \sin t)$.

$$\int_\Gamma \alpha = \int_0^{2\pi} \frac{-\sin t(-\sin t) + \cos t \cdot \cos t}{1}\,dt = \int_0^{2\pi} 1\,dt = 2\pi \neq 0$$

Since α is closed (check: $d\alpha = 0$) but $\oint \alpha \neq 0$, α is not exact, so the domain is not simply connected. □

Chapter 3: Multilinear Algebra

3.1. $B(X,Y) = \mathrm{Tr}(XY)$.
Symmetry: $\mathrm{Tr}(XY) = \sum_{i,j} X_{ij}Y_{ji} = \sum_{i,j} Y_{ji}X_{ij} = \mathrm{Tr}(YX)$.
Rank and Index: $Q(X) = \mathrm{Tr}(X^2) = \sum_{i,j} X_{ij}X_{ji}$.
For diagonal entries: $\sum_i X_{ii}^2$ (all positive contributions).
For off-diagonal: $\sum_{i<j} 2X_{ij}X_{ji}$. Using the identity $(X_{ij} + X_{ji})^2 - (X_{ij} - X_{ji})^2 = 4X_{ij}X_{ji}$, we get equal numbers of positive and negative contributions.
Result: $n_+ = n + \binom{n}{2} = \frac{n(n+1)}{2}$, $n_- = \binom{n}{2} = \frac{n(n-1)}{2}$, rank $= n^2$. □

3.3. **(a)** Let $\dim V = n$ (odd) and ω a 2-form with matrix A. Since ω is skew-symmetric, $A^T = -A$. Then:

$$\det A = \det(A^T) = \det(-A) = (-1)^n \det A = -\det A$$

So $\det A = 0$, meaning ω is degenerate.
(b) The rank of a skew-symmetric bilinear form equals the dimension of the non-degenerate part. By (a), this must be even-dimensional (else it would be degenerate). Hence the rank is even. □

Chapter 5: Manifolds

5.1. **(a)** $SO(3)$**:** Define $F : \mathcal{M}_{3,3} \to \mathrm{Sym}(3)$ by $F(A) = A^T A$. Then $O(3) = F^{-1}(I)$.
The derivative: $dF_A(H) = A^T H + H^T A$.
For $A \in O(3)$ and any symmetric S, take $H = \frac{1}{2}AS$. Then $A^T H + H^T A = \frac{1}{2}S + \frac{1}{2}S^T = S$.
So dF_A is surjective, making $O(3)$ (and thus $SO(3)$) a 3-dimensional submanifold.
$UT(S^2)$**:** Define $G(x, y) = (x \cdot x, y \cdot y, x \cdot y)$. Then $UT(S^2) = G^{-1}(1, 1, 0)$.
$dG_{(x,y)} = \begin{pmatrix} 2x^T & 0 \\ 0 & 2y^T \\ y^T & x^T \end{pmatrix}$ has full rank when x, y are unit vectors with $x \perp y$.
(b) Define $\phi : SO(3) \to UT(S^2)$ by $\phi([v_1|v_2|v_3]) = (v_1, v_2)$.
This is well-defined since columns of an orthogonal matrix are orthonormal.
Inverse: $\phi^{-1}(x, y) = [x|y|x \times y]$. The cross product gives a unit vector orthogonal to both, with the correct orientation (determinant = 1).
Both maps are smooth (linear/polynomial), so this is a diffeomorphism. □

Chapter 6: Stokes' Theorem

6.1. The ellipse is parameterized by $\gamma(t) = (a \cos t, b \sin t)$ for $t \in [0, 2\pi]$.
Using Green's theorem with $P = x^2 - y$, $Q = x - y^2$:

$$\oint_C = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D (1 - (-1))\, dA = 2 \cdot \mathrm{Area}(D) = 2\pi ab$$

□

6.3. The work is $\int_\Gamma \omega$ where $\omega = (x^2 + y^2 + z^2)(x\, dx + y\, dy + z\, dz)$.
Note that $x\, dx + y\, dy + z\, dz = \frac{1}{2}d(x^2 + y^2 + z^2)$.
Let $u = x^2 + y^2 + z^2$. Then $\omega = u \cdot \frac{1}{2}du = \frac{1}{2}d(\frac{u^2}{2}) = d(\frac{u^2}{4})$.
So $f = \frac{1}{4}(x^2 + y^2 + z^2)^2$ is a primitive, and:

$$\text{Work} = f(B) - f(A) = \frac{1}{4}(b_1^2 + b_2^2 + b_3^2)^2 - \frac{1}{4}(a_1^2 + a_2^2 + a_3^2)^2$$

□

6.5. Let $\omega = (y + z)dx + (z + x)dy + (x + y)dz$.
Compute: $d\omega = (dy + dz) \wedge dx + (dz + dx) \wedge dy + (dx + dy) \wedge dz$.
Expanding: $= dy \wedge dx + dz \wedge dx + dz \wedge dy + dx \wedge dy + dx \wedge dz + dy \wedge dz$
$= -dx \wedge dy + dx \wedge dy - dx \wedge dz + dx \wedge dz - dy \wedge dz + dy \wedge dz = 0$.
So ω is closed. The curve lies on the plane $x + z = 1$ (since $\sin^2 t + \cos^2 t = 1$), which is simply connected.

By Poincaré lemma, $\omega = d\beta$ for some β. By Stokes: $\oint_C \omega = \int_S d\omega = 0$. □

Chapter 7: Advanced Topics

Degree of F_A. Since A is invertible, $F_A : S^2 \to S^2$ is smooth and surjective.
For a regular value $b \in S^2$, the preimage $F_A^{-1}(b)$ consists of the points $x \in S^2$ where Ax is parallel to b. Since A is invertible, there's exactly one such line direction, giving two points $\pm x_0$ on S^2.
At each preimage point, the sign of $\det(d_{x_0} F_A)$ depends on whether A preserves or reverses orientation. Since both points contribute the same sign:

$$\deg(F_A) = \operatorname{sign}(\det(A))$$

□

$\mathbb{R}^2$ minus two points. We show $\dim H^1(U) > 1$ where $U = \mathbb{R}^2 \setminus \{(\pm 1, 0)\}$.
Consider two 1-forms:

$$\alpha_1 = \frac{-(y)dx + (x-1)dy}{(x-1)^2 + y^2}, \quad \alpha_2 = \frac{-(y)dx + (x+1)dy}{(x+1)^2 + y^2}$$

Both are closed on U. Consider loops γ_1 around $(1, 0)$ only and γ_2 around $(-1, 0)$ only.
Then $\oint_{\gamma_1} \alpha_1 = 2\pi$, $\oint_{\gamma_1} \alpha_2 = 0$ (and similarly for γ_2).
So $[\alpha_1]$ and $[\alpha_2]$ are linearly independent in $H^1(U)$, giving $\dim H^1(U) \geq 2$.
But $\mathbb{R}^2 \setminus \{0\}$ has $\dim H^1 = 1$. Since cohomology is a diffeomorphism invariant, these spaces are not diffeomorphic. □

No degree 1 maps $S^2 \to T^2$. Suppose $f : S^2 \to T^2$ has degree 1. Then $f^* : H^2(T^2) \to H^2(S^2)$ is an isomorphism.
Now $H^1(T^2) = \mathbb{R}^2$ with generators $[d\phi_1], [d\phi_2]$, and $[d\phi_1] \wedge [d\phi_2] = [\eta]$ generates $H^2(T^2)$.
For $f^*[\eta] \neq 0$, we need $f^*[d\phi_1] \wedge f^*[d\phi_2] \neq 0$ in $H^2(S^2)$.
But $f^*[d\phi_i] \in H^1(S^2) = 0$ (since S^2 is simply connected).
So $f^*[d\phi_i] = 0$, hence $f^*[\eta] = 0$, contradicting degree 1. □

Linking number. The linking number $\operatorname{lk}(\phi, \psi)$ counts signed intersections of ψ with any disk bounded by ϕ.
ϕ bounds the disk $D = \{(x, y, 0) : x^2 + y^2 \leq 1\}$.
ψ intersects D when $\gamma_2(t) = 0$ (the z-coordinate is 0) and $|\gamma_1(t)| < 1$ (inside the unit disk).
The sign at each intersection is $\operatorname{sign}(\gamma_2'(t))$ (positive if crossing upward).
The winding number around $(-1, 0)$ counts signed crossings of the negative x-axis by γ, which is the sum of signs for $\gamma_1 < -1$.
The winding number around $(1, 0)$ counts signed crossings for $\gamma_1 > 1$.
The linking number counts crossings for $-1 < \gamma_1 < 1$:

$$\operatorname{lk}(\phi, \psi) = w(\gamma, (-1, 0)) - w(\gamma, (1, 0)) = 2 - (-3) = 5$$

□